

$$8a - 9 = 15 \Rightarrow a = 3$$

A01/4/193. Find the integral value of k for which the system of equation

$$\log_x(y^2) + \log_y(x^5) = 2k - 1$$

$$\log_{(x^2)}(y^5) - \log_{(y^2)}(x^3) = k - 3 \text{ has real solutions } \underline{\hspace{2cm}}.$$

k का पूर्णांक मान ज्ञात कीजिए जबकि समीकरण का निकाय

$$\log_x(y^2) + \log_y(x^5) = 2k - 1$$

$$\log_{(x^2)}(y^5) - \log_{(y^2)}(x^3) = k - 3 \text{ के वास्तविक हल } \underline{\hspace{2cm}} \text{ होगा।}$$

Ans. 4

Sol. Let $\log_x y = t$, the equations become

$$2t + \frac{5}{t} = 2k - 1 \quad \underline{\hspace{2cm}} (1)$$

$$\frac{5t}{2} - \frac{3}{2t} = k - 3 \quad \underline{\hspace{2cm}} (2)$$

Substituting k from (2) in (1)

$$2t + \frac{5}{t} = 2\left(\frac{5t}{2} - \frac{3}{2t} + 3\right) - 1 - 1$$

$$\Rightarrow 3t^2 + 5t - 8 = 0 \quad \Rightarrow t = 1, -\frac{8}{3}$$

$$\Rightarrow k = 4 \text{ or } \frac{-149}{48} \quad \Rightarrow k = 4$$

A01/4/194. Let $P = x^3 - \frac{1}{x^3}$ and $Q = x - \frac{1}{x}$ where $Q \in \mathbb{R}^+$, if the minimum value of $\frac{P}{Q^2}$ is $k\sqrt{3}$, then the value of 'k' is .

यदि $P = x^3 - \frac{1}{x^3}$ और $Q = x - \frac{1}{x}$, जहाँ $Q \in \mathbb{R}^+$ है, तो यदि $\frac{P}{Q^2}$ का न्यूनतम मान $k\sqrt{3}$ है, तो 'k' का मान होगा।

Ans. 2

Sol. $\therefore Q = x - \frac{1}{x}$

$$\Rightarrow Q^3 = x^3 - \frac{1}{x^3} - 3 \cdot x \cdot \frac{1}{x} \left(x - \frac{1}{x} \right)$$

$$\Rightarrow Q^3 = P - 3Q \quad \Rightarrow P = Q^3 + 3Q$$

$$\Rightarrow \frac{P}{Q^2} = Q + \frac{3}{Q} \text{ ————— (i)}$$

$$\text{Now, } \frac{Q + \frac{3}{Q}}{2} \geq \sqrt{Q \cdot \frac{3}{Q}} \quad \Rightarrow \quad Q + \frac{3}{Q} \geq 2\sqrt{3}$$

$$\Rightarrow \frac{P}{Q^2} \geq 2\sqrt{3} \quad \therefore k = 2$$

A01/4/195. If α, β are roots of the equation $x^2 - 34x + 1 = 0$, then the value of $\left| \sqrt[4]{\alpha} - \sqrt[4]{\beta} \right|$ is (where $\sqrt[4]{}$ denotes the principal value) _____.

यदि α, β समीकरण $x^2 - 34x + 1 = 0$ के मूल हैं, तो $\left| \sqrt[4]{\alpha} - \sqrt[4]{\beta} \right|$ का मान _____ होगा। (जहाँ $\sqrt[4]{}$ प्रमुख मान को निरूपित करता है)

Ans. 2

Sol. $\therefore \alpha, \beta$ be the roots of the equation $x^2 - 34x + 1 = 0 \quad \therefore \alpha + \beta = 34, \alpha\beta = 1$

$$\text{Now, } \left(\alpha^{\frac{1}{4}} - \beta^{\frac{1}{4}} \right)^2 = \sqrt{\alpha} + \sqrt{\beta} - 2(\alpha\beta)^{\frac{1}{4}}$$

$$\Rightarrow \left(\alpha^{\frac{1}{4}} - \beta^{\frac{1}{4}} \right)^2 = \sqrt{\alpha} + \sqrt{\beta} - 2 \text{ ————— (i) } (\because \alpha\beta = 1)$$

$$\text{Now, } \left(\sqrt{\alpha} + \sqrt{\beta} \right)^2 = \alpha + \beta + 2\sqrt{\alpha\beta}$$

$$\Rightarrow \left(\sqrt{\alpha} + \sqrt{\beta} \right)^2 = 36 \quad (\because \alpha + \beta = 34, \alpha\beta = 1)$$

$$\therefore \sqrt{\alpha} + \sqrt{\beta} = 6 \quad (\because \text{we consider the principal value})$$

$$\therefore \text{From (i); } \left(\alpha^{\frac{1}{4}} - \beta^{\frac{1}{4}} \right)^2 = 6 - 2 = 4$$

$$\therefore \alpha^{\frac{1}{4}} - \beta^{\frac{1}{4}} = \pm 2$$

A02/4/196. a, b and c are all different and non-zero real numbers in arithmetic progression. If the roots of quadratic

equation $ax^2 + bx + c = 0$ are α and β such that $\frac{1}{\alpha} + \frac{1}{\beta}$, $\alpha + \beta$ and $\alpha^2 + \beta^2$ are in geometric progression,

then the value of $\frac{a}{c}$ will be _____.

a, b तथा c सभी भिन्न वास्तविक अशून्य संख्यायें समान्तर श्रेणी में हैं। यदि α तथा β समीकरण $ax^2 + bx + c = 0$ के मूल इस प्रकार हैं कि $\frac{1}{\alpha} + \frac{1}{\beta}$, $\alpha + \beta$ तथा $\alpha^2 + \beta^2$ गुणोत्तर श्रेणी में हैं तो $\frac{a}{c}$ का मान _____ होगा।

Ans. 3

Sol. $(\alpha + \beta)^2 = \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)(\alpha + \beta)^2$

$$\Rightarrow (\alpha + \beta)^2 = \frac{(\alpha + \beta)}{\alpha\beta} \{(\alpha + \beta)^2 - 2\alpha\beta\}$$

Substituting $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$ we have

$$\frac{b^2}{a^2} = \frac{-b}{c} \left(\frac{b^2}{a^2} - \frac{2c}{a} \right)$$

$$\Rightarrow cb^2 + b(b^2 - 2ac) = 0$$

$$\because b \neq 0 \therefore bc + b^2 - 2ac = 0$$

$$\because a, b, c \text{ are in AP } \therefore b = \frac{a+c}{2}$$

$$\therefore \text{we have } \frac{(a+c)c}{2} + \left(\frac{a+c}{2}\right)^2 - 2ac = 0$$

$$\Rightarrow a^2 - 4ac + 3c^2 = 0 \quad \Rightarrow (a-c)(a-3c) = 0$$

$$\because a \neq c \therefore a = 3c \quad \therefore \frac{a}{c} = 3$$

A02/4/197. Find the sum of the series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{12} + \dots \infty$, where the terms are the reciprocals of the positive integers whose only prime factors are two's and three's _____.

श्रेणी का योग ज्ञात कीजिए $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{12} + \dots \infty$, जहाँ पद उन धनात्मक पूर्णाकों के व्युत्क्रम हैं जिनके केवल अभाज्य गुणनखंड दो और तीन हैं _____।

Ans. 3

Sol. $3^0 \{2^0 + 2^{-1} + 2^{-2} \dots \infty\} = 1 \{2\}$

$$3^{-1}\{2^0 + 2^{-1} + 2^{-2} \dots \infty\} = \frac{1}{3}\{2\}$$

$$3^{-2}\{2^0 + 2^{-1} + 2^{-2} \dots \infty\} = \frac{1}{3}\{2\}$$

⋮

⋮

∞

$$\text{Hence } \frac{2 \times 1}{1 - \frac{1}{3}} = 3$$

A02/4/198. Let x be the arithmetic mean and y, z be the two geometric means between any two positive numbers. Then $\frac{y^3 + z^3}{xyz} = \dots$

माना x समान्तर माध्य तथा y, z दो धनात्मक संख्याओं के मध्य दो गुणोत्तर माध्य है, तो $\frac{y^3 + z^3}{xyz} = \dots$

Ans. 2

Sol. Let no. are a, b

$$x = \frac{a+b}{2}$$

for y and z

$$y = a \left(\frac{b}{a} \right)^{1/3}$$

$$z = a \left(\frac{b}{a} \right)^{2/3}$$

$$\frac{y^3 + z^3}{xyz} = \frac{a^2b + b^2a}{\left(\frac{a+b}{2} \right) \left(a \left(\frac{b}{a} \right) \right)} = 2$$

A02/4/199. Find minimum value of $\log_{x_1} \left(x_2 - \frac{1}{4} \right) + \log_{x_2} \left(x_3 - \frac{1}{4} \right) + \log_{x_3} \left(x_4 - \frac{1}{4} \right) + \log_{x_4} \left(x_1 - \frac{1}{4} \right)$ where

$$x_i \in \left(\frac{1}{4}, 1 \right) \text{ —————.}$$

$\log_{x_1}\left(x_2 - \frac{1}{4}\right) + \log_{x_2}\left(x_3 - \frac{1}{4}\right) + \log_{x_3}\left(x_4 - \frac{1}{4}\right) + \log_{x_4}\left(x_1 - \frac{1}{4}\right)$ का न्यूनतम मान ज्ञात कीजिए, जहाँ

$$x_i \in \left(\frac{1}{4}, 1\right) \text{ —————}$$

Ans. 8

Sol. $\underbrace{\log_{x_1}\left(x_2 - \frac{1}{4}\right), \log_{x_2}\left(x_3 - \frac{1}{4}\right), \log_{x_3}\left(x_4 - \frac{1}{4}\right), \log_{x_4}\left(x_1 - \frac{1}{4}\right)}_{\text{all 4 numbers are positive}}$

AM $\geq$ G.M.

$$\begin{aligned} & \log_{x_1}\left(x_2 - \frac{1}{4}\right) + \log_{x_2}\left(x_3 - \frac{1}{4}\right) + \log_{x_3}\left(x_4 - \frac{1}{4}\right) + \log_{x_4}\left(x_1 - \frac{1}{4}\right) \\ & \geq 4 \left(\log_{x_1}\left(x_2 - \frac{1}{4}\right) \cdot \log_{x_2}\left(x_3 - \frac{1}{4}\right) \cdot \log_{x_3}\left(x_4 - \frac{1}{4}\right) \cdot \log_{x_4}\left(x_1 - \frac{1}{4}\right) \right)^{1/4} \\ & = 4 \left(\log_{x_1}\left(x_1 - \frac{1}{4}\right) \cdot \log_{x_2}\left(x_2 - \frac{1}{4}\right) \cdot \log_{x_3}\left(x_3 - \frac{1}{4}\right) \cdot \log_{x_4}\left(x_4 - \frac{1}{4}\right) \right)^{1/4} \\ & \geq 4(2 \cdot 2 \cdot 2 \cdot 2)^{1/4} \geq 8 \end{aligned}$$

$$\therefore \text{We have } \left(x_1 - \frac{1}{2}\right)^2 \geq 0$$

$$x_1^2 - x_1 + \frac{1}{4} \geq 0$$

$$x_1^2 \geq x_1 - \frac{1}{4}$$

$$\log_{x_1} x_1^2 \leq \log_{x_1} \left(x_1 - \frac{1}{4}\right) \quad (\because x_1 < 1)$$

$$\log_{x_1} \left(x_1 - \frac{1}{4}\right) \geq 2$$

A02/4/200. If $x, y, z > 0$ and $x + y + z = 1$, then find the least value of $\frac{2x}{1-x} + \frac{2y}{1-y} + \frac{2z}{1-z}$ —————.

यदि $x, y, z > 0$ और $x + y + z = 1$ हो, तो $\frac{2x}{1-x} + \frac{2y}{1-y} + \frac{2z}{1-z}$ का न्यूनतम मान _____ होगा।

Ans. 3

Sol. $\because x, y, z > 0 \Rightarrow A.M. \geq H.M.$

$$\Rightarrow \frac{(x+y) + (y+z) + (z+x)}{3} \geq \frac{3}{\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x}}$$

$$\Rightarrow 2(x+y+z) \left(\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x} \right) \geq 9$$

$$\Rightarrow 2 \left(\frac{x+y+z}{x+y} + \frac{x+y+z}{y+z} + \frac{x+y+z}{z+x} \right) \geq 9$$

$$\Rightarrow 2 \left[\left(1 + \frac{z}{x+y} \right) + \left(1 + \frac{x}{y+z} \right) + \left(1 + \frac{y}{x+z} \right) \right] \geq 9 \quad \Rightarrow 2 \left(\frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y} \right) \geq 3$$

$$\Rightarrow 2 \left(\frac{x}{1-x} + \frac{y}{1-y} + \frac{z}{1-z} \right) \geq 3 \quad \{ \because x+y+z=1 \}$$

A02/4/201. Consider the infinite series $1 + 3x + 7x^2 + 15x^3 + \dots + (1 + 2 + 2^2 + \dots + 2^n)x^n + \dots$ whose sum is defined if $-\frac{1}{2} < x < \frac{1}{2}$. If the value of the sum is $\frac{8}{3}$ when $x = x_0$ then find the value of $24x_0$ is _____.

अनंत श्रेणी $1 + 3x + 7x^2 + 15x^3 + \dots + (1 + 2 + 2^2 + \dots + 2^n)x^n + \dots$ पर विचार कीजिए, जिसका योग निम्न शर्त के तहत परिभाषित है यदि $-\frac{1}{2} < x < \frac{1}{2}$ । यदि इस श्रेणी का योग $\frac{8}{3}$ है जब $x = x_0$ हो, तो $24x_0$ का मान _____ होगा।

Ans. 6

Sol. $S_\infty = 1 + 3x + 7x^2 + 15x^3 + 31x^4 + \dots$
 $S_\infty x = x + 3x^2 + 7x^3 + 15x^4 + 31x^5 + \dots$
 $S_\infty(1-x) = 1 + 2x + 4x^2 + 8x^3 + 16x^4 + \dots$

$$\Rightarrow S_\infty = \frac{1}{(1-x)(1-2x)} = \frac{8}{3} \Rightarrow x_0 = \frac{1}{4}$$

A02/4/202. The difference between the sum of the first k terms of the series $1^3 + 2^3 + 3^3 + \dots + n^3$ and the sum of the first k terms of $1 + 2 + 3 + \dots + n$ is 1980. The value of k is _____.

पहली k पदों की श्रेणी $1^3 + 2^3 + 3^3 + \dots + n^3$ के योग और पहली k पदों की श्रेणी $1 + 2 + 3 + \dots + n$ के योग का अंतर 1980 है। k का मान _____ है।

Ans. 9

Sol. $\left[\frac{k(k+1)}{2} \right]^2 - \frac{k(k+1)}{2} = 1980 \Rightarrow \frac{k(k+1)}{2} \left[\frac{k(k+1)}{2} - 1 \right] = 1980$

$$\Rightarrow k(k+1)(k^2+k-2) = 1980 \times 4$$

$$\Rightarrow (k-1)k(k+1)(k+2) = 8 \cdot 9 \cdot 10 \cdot 11$$

A02/4/203. If $S = \sum_{n=2}^{\infty} \frac{3n^2+1}{(n^2-1)^3}$ then $16S =$ _____.

यदि $S = \sum_{n=2}^{\infty} \frac{3n^2+1}{(n^2-1)^3}$ है तो $16S =$

Ans. 9

Sol. $(n^3-1)^3 = (n+1)^3(n-1)^3$
 $(n+1)^3 - (n-1)^3 = 6n^2 + 2$

$$\frac{3n^2+1}{(n^2-1)^3} = \frac{1}{2} \frac{6n^2+2}{(n^2-1)^3} = \frac{1}{2} \left(\frac{(n+1)^3 - (n-1)^3}{(n+1)^3(n-1)^3} \right)$$

$$= \frac{1}{2} \left(\frac{1}{(n-1)^3} - \frac{1}{(n+1)^3} \right)$$

$$S = \frac{1}{2} \left[\left(\frac{1}{1^3} - \frac{1}{3^3} \right) + \left(\frac{1}{2^3} - \frac{1}{4^3} \right) + \left(\frac{1}{3^3} - \frac{1}{5^3} \right) + \left(\frac{1}{4^3} - \frac{1}{6^3} \right) + \dots \right]$$

$$\frac{1}{2} \left(1 + \frac{1}{8} \right) = \frac{9}{18} \Rightarrow 16S = 9 \Rightarrow 16 = \frac{9}{S}$$

A02/4/204. Let $S = \sum_{n=1}^{16} \frac{n^4}{4n^2-1}$. Find the greatest integer less than or equal to S _____.

माना $S = \sum_{n=1}^{16} \frac{n^4}{4n^2-1}$ है तो S का पूर्णांक भाग _____ होगा।

Ans. 375

Sol. $t_n = \frac{n^4}{4n^2-1} = \frac{16n^4-1+1}{16(4n^2-1)} = \frac{1}{16}(4n^2+1) + \frac{1}{16(4n^2-1)}$

$$= \frac{1}{16}(4n^2+1) + \frac{1}{32} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right)$$

$$\sum_{n=1}^{16} t_n = \frac{1}{16} \left[\frac{4 \cdot 16 \cdot 17 \cdot 33}{6} + 16 \right] + \frac{1}{32} \left(1 - \frac{1}{33} \right)$$

$$S = 375 + \frac{1}{33} \Rightarrow [S] = 375$$

A02/4/205. If $\sum_{n=0}^{2015} \frac{1}{7^n + \sqrt{7^{2015}}} = \frac{144}{\sqrt{7^{2010+k}}}$ where k is a natural numbers. Find k _____.

यदि $\sum_{n=0}^{2015} \frac{1}{7^n + \sqrt{7^{2015}}} = \frac{144}{\sqrt{7^{2010+k}}}$ जहाँ k एक प्राकृतिक संख्या है। k _____ का मान होगा।

Ans. 3

Sol.
$$\sum_{n=0}^{2015} \frac{1}{7^n + \sqrt{7^{2015}}}$$

$$= \frac{1}{1 + \sqrt{7^{2015}}} + \frac{1}{7 + \sqrt{7^{2015}}} + \frac{1}{7^2 + \sqrt{7^{2015}}} + \dots + \frac{1}{7^{2013} + \sqrt{7^{2015}}} + \frac{1}{7^{2014} + \sqrt{7^{2015}}} + \frac{1}{7^{2015} + \sqrt{7^{2015}}}$$

$$= \frac{1008}{\sqrt{7^{2015}}} = \frac{144}{\sqrt{7^{2013}}}$$

A02/4/206. Let $a_1 = \frac{1}{2}$, $a_{k+1} = a_k^2 + a_k \forall k \geq 1$

$x_n = \sum_{k=1}^n \frac{1}{a_k + 1}$ find $[x_{100}]$ where $[\cdot]$ is greatest integer function _____.

माना कि $a_1 = \frac{1}{2}$, $a_{k+1} = a_k^2 + a_k \forall k \geq 1$

$x_n = \sum_{k=1}^n \frac{1}{a_k + 1}$ जहाँ $[\cdot]$ महत्तम पूर्णांक फलन है, $[x_{100}]$ का मान ज्ञात कीजिए।

Ans. 1

Sol. $a_{k+1} = a_k(a_k + 1)$

$$\frac{1}{a_k + 1} = \frac{1}{a_k} - \frac{1}{(a_k) + 1}$$

$$\frac{1}{(a_k) + 1} = \frac{1}{a_k} - \frac{1}{a_{k+1}}$$

$$x_n = \frac{1}{a_1} - \frac{1}{a_{n+1}} = 2 - \frac{1}{a_{n+1}}$$

As $a_{101} > 1$, we have $[x_{100}] = 1$

A02/4/207. Let $\log_2 x$, $(1 + \log_4 x)$ and $(\log_8 4x)$ are three consecutive terms of G.P with common ratio $r(r > 0)$. If α, β are the roots of the quadratic equation $(a - a^2 - 1)x^2 + 2ax + 1 = 0$ such that $\beta < 3r < \alpha$ then the integral value of a is _____.

माना $\log_2 x$, $(1 + \log_4 x)$ तथा $(\log_8 4x)$ सार्वअनुपात $r(r > 0)$ वाली गुणोत्तर श्रेणी के तीन क्रमागत पद हैं। यदि α, β द्विघात

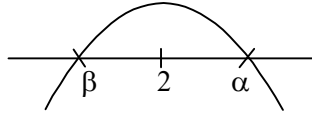
समीकरण $(a - a^2 - 1)x^2 + 2ax + 1 = 0$ के मूल इस प्रकार हैं कि $\beta < 3r < \alpha$ हो, तो a का पूर्णांक मान _____ होगा।

Ans. 1

Sol.
$$r = \frac{(2+t)}{2t} = \frac{(2+t).2}{3.(2+t)} \Rightarrow \boxed{r = \frac{2}{3}}$$

where $t = \log_2 x$

$$\therefore 3r = 2$$



Let $f(x) = (a - a^2 - 1)x^2 + 2ax + 1$

$\therefore$ required condition is $f(2) > 0$

$$\Rightarrow 4a^2 - 8a + 3 < 0$$

$$\Rightarrow (2a - 1)(2a - 3) < 0$$

$$\Rightarrow \frac{1}{2} < a < \frac{3}{2}$$

$\therefore$ integral value of $a = 1$

A02/4/208. The terms of an arithmetic sequence add to 715. The first term of sequence is increased by 1, the second term is increased by 3 and so on, in general, the k th term is increased by the k th odd positive integer. The terms of new sequence add to 836. If a_1 and a_n denotes first and last term of the original sequence, then

$$\left(\frac{a_1 + a_n}{65} \right) \text{ is } \underline{\hspace{2cm}}.$$

किसी समान्तर श्रेणी के पदों का योगफल 715 है। श्रेणी के प्रथम पद में 1, द्वितीय पद में 3 तथा इसी प्रकार आगे पदों में भी अंक जोड़े जाते हैं अर्थात् k वे पद में k वीं विषम धनात्मक पूर्णांक जोड़ा जाता है। नई श्रेणी के सभी पदों का योगफल 836 है। यदि a_1

तथा a_n मूल श्रेणी के प्रथम तथा अन्तिम पद को दर्शाते हो तो $\left(\frac{a_1 + a_n}{65} \right)$ का मान _____ होगा।

Ans. 2

Sol. Given $a_1 + a_2 + \dots + a_n = 715 = \frac{n}{2}(a_1 + a_n)$ _____ (1)

and

$$(a_1 + 1) + (a_2 + 3) + \dots + (a_n + (2n - 1)) = 836$$
 _____ (2)

$$\text{on (2) - (1)} \Rightarrow \sum_{n=1}^n (2n - 1) = 121$$

$$\Rightarrow n^2 = 121$$

$$\Rightarrow n = 11$$

$$\text{Put value of } n \text{ in equation (1), } (a_1 + a_n) \cdot \frac{11}{2} = 715$$

$$\Rightarrow \frac{a_1 + a_n}{65} = 2$$

A02/4/209. If a, b, c and d are positive integers and $a < b < c < d$ such that a, b, c are in A.P. and b, c, d are in G.P. and $d - a = 30$. Find the value of sum of digits of $\frac{a + b + c + d}{3}$ _____.

यदि a, b, c और d धनात्मक पूर्णांक हैं तथा $a < b < c < d$ जबकि a, b, c समान्तर श्रेणी में हैं तथा b, c, d गुणोत्तर श्रेणी में हैं तथा $d - a = 30$ तब संख्या $\frac{a + b + c + d}{3}$ के अंकों का योगफल _____ होगा।

Ans. 7

Sol. Let the numbers be माना कि संख्याएं

$$\begin{array}{ccccccc} & & \overbrace{\hspace{1.5cm}}^{\text{G.P.}} & & & & \\ A - D, & A, & A + D, & \frac{(A + D)^2}{A} & & & \\ \underbrace{\hspace{1.5cm}}_{\text{A.P.}} & \underbrace{\hspace{0.5cm}}_{(a)} & \underbrace{\hspace{0.5cm}}_{(b)} & \underbrace{\hspace{0.5cm}}_{(c)} & \underbrace{\hspace{0.5cm}}_{(d)} & & \end{array}$$

Given दिया गया है $d - a = 30$

$$\Rightarrow \frac{(A + D)^2}{A} - (A - D) = 30 \quad \Rightarrow (A + D)^2 - A(A - D) = 30A$$

$$\Rightarrow D^2 + 3AD = 30A$$

$$D^2 = 3A(10 - D)$$

$$\therefore A = \frac{D^2}{3(10 - D)} \quad \text{_____ (1)}$$

since चूंकि 'A' is a + ve integer

$$\therefore 0 < D < 10 \quad \text{_____ (2)}$$

Also since '3' is prime and A is an integer

चूंकि 3, अभाज्य है तथा A पूर्णांक है।

$\therefore D^2$ must be divisible 3

D^2 , 3 से विभाजित होगा।

$\Rightarrow D$ must be of the form of 3K

D, 3k का रूप होगा।

$\therefore$ possible values of D are 3, 6, 9

D के संभावित मान 3, 6, 9

$$D = 3 \quad \Rightarrow A = \frac{3}{7} \text{ (rejected, अस्वीकार्य)}$$

$$D = 6 \quad \Rightarrow A = 3 \text{ (rejected, अस्वीकार्य)}$$

$$D = 9 \Rightarrow A = 27$$

$\Rightarrow$ Numbers are संख्याएं हैं 18, 27, 36, 48

A02/4/110. Let $p(x) = x^2 - x + 1$. Let α be a root of $p(p(p(p(x))))$. Find the value of $|(p(\alpha) - 1) p(\alpha) p(p(\alpha)) p(p(p(\alpha)))|$ _____.

माना $p(x) = x^2 - x + 1$. माना α , $p(p(p(p(x))))$ का मूल है तब $|(p(\alpha) - 1) p(\alpha) p(p(\alpha)) p(p(p(\alpha)))|$ का मान _____ होगा।

Ans. 1

Sol. Since चूंकि $(x - 1)x = p(x) - 1$, we can set है तब

$$\begin{aligned} & (p(\alpha) - 1)p(\alpha)p(p(\alpha))p(p(p(\alpha))) \\ &= (p(p(p(\alpha))) - 1) p(p(\alpha)) p(p(p(\alpha))) \\ &= p(p(p(\alpha))) - 1 = p(p(p(p(p(\alpha)))) - 1 = -1 \end{aligned}$$

A02/4/211. Find the value of $\left(1 - \frac{\cos 61^\circ}{\cos 1^\circ}\right) \cdot \left(1 - \frac{\cos 62^\circ}{\cos 2^\circ}\right) \cdot \left(1 - \frac{\cos 63^\circ}{\cos 3^\circ}\right)$ _____.

$$\left(1 - \frac{\cos 61^\circ}{\cos 1^\circ}\right) \cdot \left(1 - \frac{\cos 62^\circ}{\cos 2^\circ}\right) \cdot \left(1 - \frac{\cos 63^\circ}{\cos 3^\circ}\right) \text{ का मान } \text{_____} \text{ होगा।}$$

Ans. 1

$$\text{Sol. } \therefore \cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \cdot \sin\left(\frac{A-B}{2}\right)$$

$$\therefore 1 - \frac{\cos(60^\circ + k)}{\cos k} = \frac{\cos k - \cos(60^\circ + k)}{\cos k} = \frac{\sin(30^\circ + k)}{\cos k}$$

$$\Rightarrow \left(1 - \frac{\cos 61^\circ}{\cos 1^\circ}\right) \left(1 - \frac{\cos 62^\circ}{\cos 2^\circ}\right) \dots \dots \dots \left(1 - \frac{\cos 119^\circ}{\cos 59^\circ}\right)$$

$$= \prod_{k=1}^{59} \frac{\sin(30^\circ + k)}{\cos k} = \prod_{k=1}^{59} \frac{\cos(60^\circ - k)}{\cos k} = 1$$

A02/4/212. $1 + \frac{(1+2)(1+3)}{2^2(1^3+2^3)} + \frac{(1+2+3)(1+3+5)}{3^2(1^3+2^3+3^3)} + \frac{(1+2+3+4)(1+3+5+7)}{4^2(1^3+2^3+3^3+4^3)} + \dots$ upto 10 terms $= \frac{a}{b}$ (where a & b are relatively prime), then a - b is equal to _____.

$$1 + \frac{(1+2)(1+3)}{2^2(1^3+2^3)} + \frac{(1+2+3)(1+3+5)}{3^2(1^3+2^3+3^3)} + \frac{(1+2+3+4)(1+3+5+7)}{4^2(1^3+2^3+3^3+4^3)} + \dots \text{ दस पदों तक} = \frac{a}{b} \text{ (जहाँ a \& b परस्पर सहभाज्य है), तो a - b का मान } \text{_____} \text{ बराबर होगा।}$$

Ans. 9

Sol.

$$T_r = \frac{(1+2+3+\dots+r)(1+3+\dots+(2r-1))}{r^2(1^3+2^3+\dots+r^3)}$$

$$= \frac{\frac{r(r+1)}{2} \cdot r^2}{r^2 \cdot \frac{r^2(r+1)^2}{4}} = \frac{2}{r(r+1)} = 2 = \left[\frac{1}{r} - \frac{1}{r+1} \right]$$

$$\therefore \sum_{r=1}^{10} T_r = 2 \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{10} - \frac{1}{11} \right)$$

$$= 2 \left(1 - \frac{1}{11} \right) = \frac{20}{11}$$

A02/4/213. If a, b, c are distinct real numbers in H.P. such that, $\left(\frac{a+b}{2a-b} \right) + \left(\frac{c+b}{2c-b} \right) > \lambda$, then the value of λ _____.

यदि a, b, c तीन विभिन्न वास्तविक संख्याएँ हरात्मक श्रेणी (H.P.) में हों, और यदि $\left(\frac{a+b}{2a-b} \right) + \left(\frac{c+b}{2c-b} \right) > \lambda$, तो λ का मान _____ होगा।

Ans. 4

Sol. It is given that a, b and c are in H.P.

$$b = \frac{2ac}{a+c}$$

$$\text{Now, } \left(\frac{a+b}{2a-b} \right) + \left(\frac{c+b}{2c-b} \right) = \left(\frac{1+\frac{b}{a}}{2-\frac{b}{a}} \right) + \left(\frac{1+\frac{b}{c}}{2-\frac{b}{c}} \right)$$

$$= \left(\frac{1+\frac{2c}{a+c}}{2-\frac{2c}{a+c}} \right) + \left(\frac{1+\frac{2a}{c+a}}{2-\frac{2a}{c+a}} \right) = \left(\frac{a+c+2c}{2a+2c-2c} \right) + \left(\frac{a+c+2a}{2a+2c-2a} \right)$$

$$= \left(\frac{a+3c}{2a} \right) + \left(\frac{3a+c}{2c} \right) = \frac{ac+3c^2+3a^2+ac}{2ac} = \frac{3c^2+3a^2+2ac}{2ac}$$

$$= \frac{3(a^2+c^2)}{2ac} + 1 = \frac{3}{2} \left(\frac{a}{c} + \frac{c}{a} \right) + 1 \geq \frac{3}{2} \times 2 + 1 = 4$$

Hence, the value of λ is 4.

A03/4/214. If $x, y, z \in \mathbb{R}$ and $\begin{vmatrix} x & y^2 & z^3 \\ x^4 & y^5 & z^6 \\ x^7 & y^8 & z^9 \end{vmatrix} = 2$ then find the value of

$$\begin{vmatrix} y^5 z^6 (z^3 - y^3) & x^4 z^6 (x^3 - z^3) & x^4 y^5 (y^3 - x^3) \\ y^2 z^3 (y^6 - z^6) & x z^3 (z^6 - x^6) & x y^2 (x^6 - y^6) \\ y^2 z^3 (z^3 - y^3) & x z^3 (x^3 - z^3) & x y^2 (y^3 - x^3) \end{vmatrix} :$$

यदि $x, y, z \in \mathbb{R}$ तथा $\begin{vmatrix} x & y^2 & z^3 \\ x^4 & y^5 & z^6 \\ x^7 & y^8 & z^9 \end{vmatrix} = 2$ हो, तो सारणिक $\begin{vmatrix} y^5 z^6 (z^3 - y^3) & x^4 z^6 (x^3 - z^3) & x^4 y^5 (y^3 - x^3) \\ y^2 z^3 (y^6 - z^6) & x z^3 (z^6 - x^6) & x y^2 (x^6 - y^6) \\ y^2 z^3 (z^3 - y^3) & x z^3 (x^3 - z^3) & x y^2 (y^3 - x^3) \end{vmatrix}$ का

मान होगा :

Ans. 4

Sol.

A03/4/215. The minimum value of determinant $\Delta = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ -1 & -\cos \theta & 2 \end{vmatrix} \forall \theta \in \mathbb{R}$ is :

सारणिक $\Delta = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ -1 & -\cos \theta & 2 \end{vmatrix} \forall \theta \in \mathbb{R}$ का न्यूनतम मान होगा :

Ans. 3

Sol. $\Delta = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\cos \theta & 1 & \cos \theta \\ -1 & -\cos \theta & 2 \end{vmatrix} = 3(1 + \cos^2 \theta)$

its minimum value = 3

A03/4/216. If the system of equation & ${}^{n+1}C_2 \cdot x + {}^{n+1}C_3 \cdot y + 10 \cdot z = 0$, ${}^{n+1}C_3 \cdot x + 10 \cdot y + {}^{n+1}C_2 \cdot z = 0$ and $10 x^{n+1} C_2 y + {}^{n+1}C_3 \cdot z = 0$ has a non-trivial solution then find the value of n ($n \in \mathbb{N}, n \geq 2$) :

यदि समीकरणों ${}^{n+1}C_2 \cdot x + {}^{n+1}C_3 \cdot y + 10 \cdot z = 0$, ${}^{n+1}C_3 \cdot x + 10 \cdot y + {}^{n+1}C_2 \cdot z = 0$ और $10 x^{n+1} C_2 y + {}^{n+1}C_3 \cdot z = 0$ का निकाय अशून्य हल रखता है तब n ($n \in \mathbb{N}, n \geq 2$) का मान _____ होगा ।

Ans. 4

Sol. For non - trivial solution

$$\begin{vmatrix} {}^{n+1}C_2 & {}^{n+1}C_3 & 10 \\ {}^{n+1}C_3 & 10 & {}^{n+1}C_2 \\ 10 & {}^{n+1}C_2 & {}^{n+1}C_3 \end{vmatrix} = 0$$

Which is possible when

$$\underbrace{{}^{n+1}C_2 + {}^{n+1}C_3 + 10 = 0}_{\text{No solution}} \quad \text{or} \quad \underbrace{{}^{n+1}C_2 = {}^{n+1}C_3 = 10}_{n = 4}$$

$$\therefore n = 4$$

A04/4/217. If the sum of all the odd divisors of $10!$, which are of the form $3n + 1$ is S_1 and which are of the form $3n - 1$ is S_2 . Then the remainder obtained by dividing S_1 by S_2 is _____.

यदि $10!$ के सभी विषय भाजकों में से $3n + 1$ के रूप में वाले भाजकों का योग S_1 है तथा $3n - 1$ के रूप में वाले भाजकों का योग S_2 है तो S_1 को S_2 से विभाजित करने पर प्राप्त शेषफल _____ है।

Ans. 8

Sol. $\because 10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$ and
 $\because$ the divisors are odd and not a multiple of 3
 $\therefore$ the number of such divisors $= (2 + 1) \cdot (1 + 1)$
 $= 6$

and they are 1, 4, 7, 25, 35, 175

out of them 5, 35 are of type $3n - 1$ and 1, 7, 25, 175 are of type $3n + 1$

$$\Rightarrow S_1 = 208, S_2 = 40$$

$\Rightarrow$ remainder obtained by dividing S_1 by S_2 is 8.

A04/4/218. Find the number of intersection points of diagonals of 2009 sides regular polygon, which lied inside the polygon _____.

2009 भुजाओं वाले सम बहुभुज के विकर्णों के प्रतिच्छेद बिन्दुओं की संख्या ज्ञात कीजिए जब बहुभुज के अंदर स्थित है _____।

Ans. ${}^{2009}C_4$

Sol. We know that in odd sides polygon no two or more than two diagonals are parallel, so if we take any 4 vertices, we get one point of intersection of diagonals, Hence required no. of points will be ${}^{2009}C_4$.

A04/4/219. Let the number of arrangements of all the digits of the number 12345 such that atleast 3 digits will not come in its original position is N. Then the unit digit of N is _____.

संख्या 12345 के सभी अंकों के क्रमों की संख्या इस प्रकार है कि कम से कम 3 अंक इसकी मूल स्थिति में नहीं आएंगे, N का इकाई अंक है _____

Ans. 9

Sol. ${}^5C_2 D_3 + {}^5C_1 D_4 + D_5$

$${}^5C_2 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!}\right) + {}^5C_1 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right) + 5! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right)$$

$$= 10 \times 2 + 5 \times 9 + 44 = 20 + 45 + 44 = 109 \quad \therefore \text{unit digit is 9.}$$

A04/4/220. The number of ways of arranging 11 objects A, B, C, D, E, F, $\alpha, \alpha, \alpha, \beta, \beta$ so that every β lie between two α (not necessarily adjacent) is $K \times 6! \times {}^{11}C_5$, then K is _____.

11 वस्तुओं A, B, C, D, E, F, $\alpha, \alpha, \alpha, \beta, \beta$ को व्यवस्थित करने के तरीकों की संख्या ताकि प्रत्येक β दो α के बीच स्थित हो (जरूरी नहीं कि नजदीक हो) $K \times 6! \times {}^{11}C_5$ है, तो K का मान _____ होगा।

Ans. 3

Sol. There are three major ways $\alpha\alpha\beta\beta\alpha$, $\alpha\beta\beta\alpha\alpha$ and $\alpha\beta\alpha\beta\alpha$

Each major way has six empty spaces. The number of ways of putting letters at these empty spaces must be non-negative integer function of $x_1 + x_2 + \dots + x_6 = 6 = {}^{6+6-1}C_{6-1} = {}^{11}C_5$

No. of arrangements is $= 3 \times {}^{11}C_5 \times 6! \Rightarrow K = 3$

A04/4/221. The number of ordered pairs (m, n); $m, n \in \{1, 2, 3, \dots, 20\}$ such that $3^m + 7^n$ is a multiple of 10, is equal to _____.

क्रमित युग्मों (m, n); $m, n \in \{1, 2, 3, \dots, 20\}$ की संख्या, यदि $3^m + 7^n$, 10 का गुणज है, बराबर _____ होगी।

Ans. 100

Sol. End digits in power of 3 and 7 repeat in cycle of 4

$$3^m \rightarrow 3^{4k} = 1, 3^{4k+1} = 3, 3^{4k+2} = 9, 3^{4k+3} = 7$$

$$7^n \rightarrow 7^{4k} = 1, 7^{4k+1} = 7, 7^{4k+2} = 9, 7^{4k+3} = 3$$

So the number $\{1, 2, 3, \dots, 20\}$ can be divided into 4 sets

$$4k \text{ type} \quad \{4, 8, 12, 16, 20\} = A$$

$$4k + 1 \text{ type} \quad \{1, 5, 9, 13, 17\} = B$$

$$4k + 2 \text{ type} \quad \{2, 6, 10, 14, 18\} = C$$

$$4k + 3 \text{ type} \quad \{3, 7, 11, 15, 19\} = D$$

Now $3^m + 7^n$ divisible by 10, if end digit of the number is 0.

Thus if $m \in A$ then $n \in C$, if $m \in B$ then $n \in B$

If $m \in C$ then $n \in A$, if $m \in D$ then $n \in D$

So, desired number of ordered pairs (m, n)

$$= 5 \times 5 + 5 \times 5 + 5 \times 5 + 5 \times 5 = 100$$

A05/4/222. What will be the remainder after dividing $\sum_{r=1}^{10} ({}^{10}C_r)(r-5)^3$ by 10

$\sum_{r=1}^{10} ({}^{10}C_r)(r-5)^3$ को 10 से भाग देने पर, शेषफल होगा _____

Ans. 5

Sol.

$$\text{Let } \sum_{r=0}^{10} {}^{10}C_r \cdot (r-5)^3$$

replace r by 10 - r

$$\Rightarrow \sum_{r=10}^0 {}^{10}C_{10-r} \cdot (10-r-5)^3 = S$$

$$\Rightarrow \sum_{r=0}^{10} {}^{10}C_r (5-r)^3 = S$$

$$\Rightarrow 2S = 0 \quad \Rightarrow S = 0$$

$$\Rightarrow \sum_{r=0}^{10} {}^{10}C_r (r-5)^3 = 0$$

$$\Rightarrow {}^{10}C_0 (-5)^3 + \sum_{r=1}^{10} {}^{10}C_r (r-5)^3 = 0$$

$$\Rightarrow \sum_{r=1}^{10} {}^{10}C_r (r-5)^3 = 125$$

A05/4/223. The value ${}^nC_0 {}^{2n}C_n - {}^nC_1 {}^{2n-1}C_n + {}^nC_2 {}^{2n-2}C_n + \dots + (-1)^n {}^nC_n {}^nC_n$ is : _____.

${}^nC_0 {}^{2n}C_n - {}^nC_1 {}^{2n-1}C_n + {}^nC_2 {}^{2n-2}C_n + \dots + (-1)^n {}^nC_n {}^nC_n$ का मान _____ होगा।

Ans. 1

Sol. Note that, $C_0 {}^{2n}C_n - C_1 {}^{2n-1}C_n + C_2 {}^{2n-2}C_n - C_3 {}^{2n-3}C_n + \dots + (-1)^n {}^nC_n {}^nC_n$
= coefficient of

$$x^n \text{ in } \left[C_0(1+x)^{2n} - C_1(1+x)^{2n-1} + C_2(1+x)^{2n-2} - C_3(1+x)^{2n-3} + \dots + (-1)^n C_n(1+x)^n \right]$$

= coefficient of x^n in

$$(1+x)^n \left[C_0(1+x)^n - C_1(1+x)^{n-1} + C_2(1+x)^{n-2} - C_3(1+x)^{n-3} + \dots + (-1)^n C_n \right]$$

= coefficient of x^n in $(1+x)^n [(1+x) - 1]^n$

= coefficient of x^n in $(1+x)^n = 1$

= coefficient of the constant term in $(1+x)^n = 1$

A05/4/224. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\sum_{t=0}^{r-1} \frac{1}{5^n} \cdot {}^nC_r \cdot {}^rC_t 3^t \right)$ is equal to _____.

$\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\sum_{t=0}^{r-1} \frac{1}{5^n} \cdot {}^nC_r \cdot {}^rC_t 3^t \right)$ का मान _____ बराबर होगा।

Ans. 1

Sol.

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{5^n} {}^nC_r \left(\sum_{t=0}^{r-1} {}^rC_t 3^t \right) = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{5^n} {}^nC_r (4^r - 3^r)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{5^n} \left(\sum_{r=1}^n {}^nC_r 4^r - \sum_{r=1}^n {}^nC_r 3^r \right) = \lim_{n \rightarrow \infty} \frac{1}{5^n} (5^n - 4^n) = 1$$

A06/4/225. Let A_n and B_n be square matrices of order 3, which are defined as

$$A_n = [a_{ij}] \text{ and } B_n = [b_{ij}] \text{ where } a_{ij} = \frac{2i+j}{3^{2n}} \text{ and } b_{ij} = \frac{3i-j}{2^{2n}} \text{ for all } i \text{ and } j, 1 \leq i, j \leq 3.$$

If $\ell = \lim_{n \rightarrow \infty} \text{Tr.}(3A_1 + 3^2 A_2 + 3^3 A_3 + \dots + 3^n A_n)$ and $m = \lim_{n \rightarrow \infty} \text{Tr.}(2B_1 + 2^2 B_2 + 2^3 B_3 + \dots + 2^n B_n)$, then find the value of $(\ell + m)$. [Note : Tr.(P) denotes the trace of matrix P.] _____.

माना कि A_n और B_n क्रम 3 के वर्ग आव्यूह हैं, जिन्हें इस प्रकार परिभाषित किया गया है कि

$$A_n = [a_{ij}] \text{ और } B_n = [b_{ij}] \text{ जहाँ } a_{ij} = \frac{2i+j}{3^{2n}} \text{ और } b_{ij} = \frac{3i-j}{2^{2n}} \text{ के लिए सभी } i \text{ तथा } j, 1 \leq i, j \leq 3.$$

यदि $\ell = \lim_{n \rightarrow \infty} \text{Tr.}(3A_1 + 3^2 A_2 + 3^3 A_3 + \dots + 3^n A_n)$ and

$$m = \lim_{n \rightarrow \infty} \text{Tr.}(2B_1 + 2^2 B_2 + 2^3 B_3 + \dots + 2^n B_n),$$

$(\ell + m)$ का मान होगा। [Note : Tr.(P) आव्यूह P के ट्रेस को दर्शाता है] _____

Ans. 21

Sol.

$$\ell = \lim_{n \rightarrow \infty} 18 \left(\frac{3}{3^2} + \frac{3^2}{3^4} + \frac{3^3}{3^6} + \dots \right)$$

$$= 18 \times \frac{1}{3 \left(1 - \frac{1}{3} \right)} = 9$$

$$m = \lim_{n \rightarrow \infty} 12 \left(\frac{2}{2^2} + \frac{2^2}{2^4} + \dots \right)$$

$$= 12 \times \frac{1}{2 \left(1 - \frac{1}{2} \right)} = 12$$

$$\therefore \ell + m = 9 + 12 = 21$$

A06/4/226. Let the number of 4×4 matrices whose entries are each '2018' or '-2018' such that the sum of the entries in each row and in each column is '0' are " $15k$ ", then k equals to _____.

यदि 4×4 क्रम के आव्यूहों की संख्या " 15 " है जिनके प्रत्येक अवयव '2018' या '-2018' इस प्रकार है कि प्रत्येक पंक्ति और

प्रत्येक स्तम्भ में अवयवों का योगफल '0' है, तो k बराबर _____ होगा।

Ans. 6

Sol. 16 elements $\begin{cases} 8 \text{ "2018'S"} \\ 8 \text{ "2018's"} \end{cases}$ we only need to arrange 8 '2018's, such that there are two in each row and column. We start placing 2018's row by row, in top row, we can place 2 '2018' in 4C_2 ways
Now, consider the 3 cases:

C - I: 2 '2018's, is 2nd row each directly under a 2018, of first row

$$\begin{bmatrix} 2018 & - & 2018 & - \\ 2018 & - & 2018 & - \\ - & - & - & - \\ - & - & - & - \end{bmatrix} \text{ Then, ways to place remaining 2018's} = 1$$

C - II Exactly one '2018' of 2nd row is under '2018' of 1st row

$$\begin{bmatrix} 2018 & - & 2018 & - \\ 2018 & - & - & 2018 \\ - & 2018 & - & - \\ - & 2018 & - & - \end{bmatrix} \therefore \text{ number of arrangements of 2018's} = 2 \times {}^2C_1 \times 2 = 8$$

C - III No 2018 of 2nd row under 2018 of 1st row

$$\begin{bmatrix} 2018 & - & 2018 & - \\ - & 2018 & - & 2018 \\ - & - & - & - \\ - & - & - & - \end{bmatrix} \text{ ways to place remaining 2018} = 2[1 + 2] = 6$$

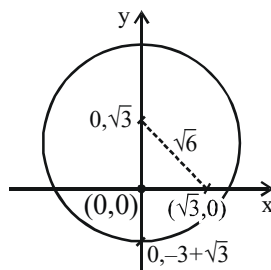
$$\therefore \text{ Total such matrices} = 6 [1 + 8 + 6] = 90$$

A07/4/227. If there exist atleast one value of the complex number $z = x + iy$, satisfy the condition $|z - \sqrt{3}| = a^2 - 4a + 3$ and the inequality $|z - \sqrt{3}i| < a^2$, then least integral value of a is _____.

यदि सम्मिश्र संख्या $z = x + iy$ का कम से कम एक मान विद्यमान है, जो प्रतिबन्ध $|z - \sqrt{3}| = a^2 - 4a + 3$ तथा असमिका $|z - \sqrt{3}i| < a^2$ को सन्तुष्ट करता है, तो a का न्यूनतम पूर्णांक मान _____ होगा।

Ans. 3

Sol. for $a = 3$,
point $(\sqrt{3}, 0)$
lies in the circle



A07/4/228. A complex number z satisfies the system of equations $|z-4|=|z-4-3i|$, $|z-2+3i|=|z-5i|$, then $2\operatorname{Re}(z)$ = _____.

एक सम्मिश्र संख्या z समीकरण निकाय $|z-4|=|z-4-3i|$, $|z-2+3i|=|z-5i|$ को सन्तुष्ट करता है तो $2\operatorname{Re}(z)$ = _____ होगा।

Ans. 6

Sol. $|z-4|=|z-4-3i|$

$$\Rightarrow z \text{ lies on } \perp \text{ bisector } y = \frac{3}{2}$$

$$|z-2+3i|=|z-5i|$$

$$\Rightarrow z \text{ lies on } \perp \text{ bisector } \Rightarrow 4y = x + 3$$

A07/4/229. If z_1 satisfies the equation $|z-3|=4$ and z_2 satisfies the equation $|z-1|+|z+1|=3$. If $m=|z_1-z_2|_{\min}$ and $M=|z_1-z_2|_{\max}$, then unit digit of $(m+2M)$ is _____.

यदि सम्मिश्र संख्या z_1 समीकरण $|z-3|=4$ को सन्तुष्ट करती है तथा z_2 समीकरण $|z-1|+|z+1|=3$ को सन्तुष्ट करती हैं। यदि $m=|z_1-z_2|_{\min}$ तथा $M=|z_1-z_2|_{\max}$ हो, तो $(m+2M)$ के इकाई अंक का मान _____ होगा।

Ans. 7

Sol. $|z_1-z_2|_{\min}=0$

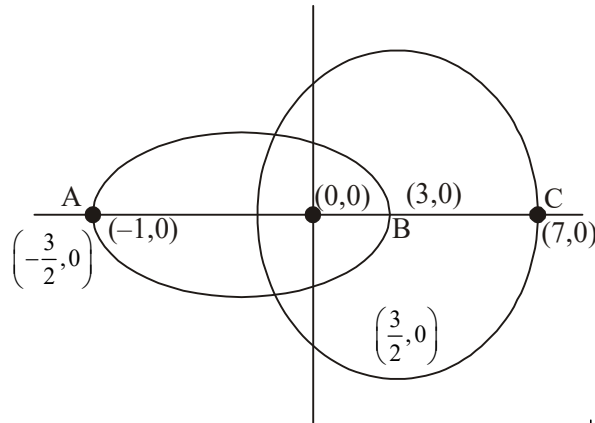
($\therefore$ curves are intersecting)

$$\Rightarrow m=0$$

$$\& |z_1-z_2|_{\max}=AC$$

$$=8+\frac{1}{2}=\frac{17}{2}=M$$

$$\therefore m+2M=17.$$



A07/4/230. If $A(z_1), B(z_2), C(z_3)$ are the vertices of an equilateral triangle ABC . If the value of $\left| \arg \left(\frac{z_2+z_3-2z_1}{2z_1-4z_2+2z_3} \right) \right|$ is $\frac{P\pi}{Q}$ (P and Q are in their lowest form), then find the value of $(P+Q)$ _____.

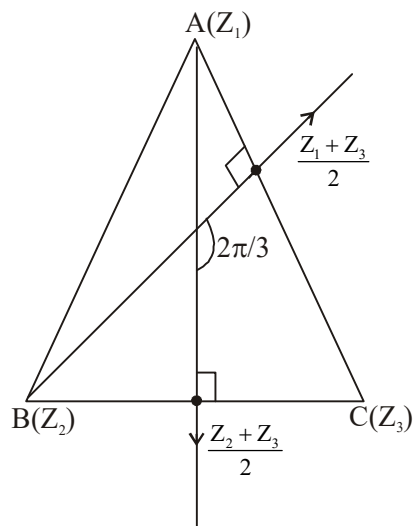
यदि $A(z_1), B(z_2), C(z_3)$ समबाहु त्रिभुज ABC के शीर्ष हैं। यदि $\left| \arg \left(\frac{z_2+z_3-2z_1}{2z_1-4z_2+2z_3} \right) \right|$ का मान $\frac{P\pi}{Q}$

(P तथा Q अपनी न्यूनतम अवस्था में) है, तो $(P+Q)$ का मान _____ होगा।

Ans. 5

Sol. $\left| \arg \left(\frac{z_2+z_3-2z_1}{2z_1-4z_2+2z_3} \right) \right|$

$$= \left| \arg \left(\frac{2 \left(\frac{z_2 + z_3}{2} - z_1 \right)}{4 \left(\frac{z_1 + z_3}{2} - z_2 \right)} \right) \right|$$



$$= \left| \arg \left[\frac{\left(\frac{z_2 + z_3}{2} - z_1 \right)}{\left(\frac{z_1 + z_3}{2} - z_2 \right)} \right] \right| = \frac{2\pi}{3}$$

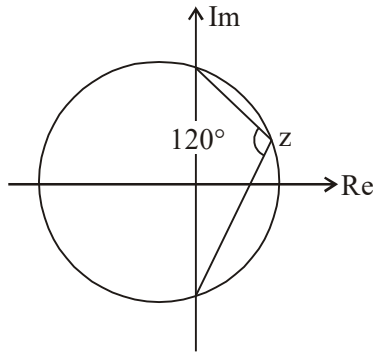
A07/4/231. If area of region enclosed by locus of z given by $\text{Arg}(z+i) - \text{Arg}(z-i) = \frac{2\pi}{3}$ and imaginary axis is $\frac{a\pi}{b} - \frac{c}{\sqrt{d}}$ (where $a, b, c, d \in \mathbb{I}$), then the value of $a + b - c - d$ is equal to _____.

यदि $\text{Arg}(z+i) - \text{Arg}(z-i) = \frac{2\pi}{3}$ को संतुष्ट करने वाले z का बिन्दुपथ तथा काल्पनिक अक्ष द्वारा परिबद्ध क्षेत्र का क्षेत्रफल

$\frac{a\pi}{b} - \frac{c}{\sqrt{d}}$ (जहाँ $a, b, c, d \in \mathbb{I}$) हो, तो $a + b - c - d$ का मान _____ होगा।

Ans. 9

Sol. Locus is are of circle with center $\left(-\frac{1}{\sqrt{3}}, 0\right)$ and radius $\frac{2}{\sqrt{3}}$



$$\text{Area} = \frac{1}{3} \times \pi \left(\frac{2}{\sqrt{3}} \right)^2 - \frac{1}{2} \times 2 \times \frac{1}{\sqrt{3}} = \frac{4\pi}{9} - \frac{1}{\sqrt{3}}$$

A07/4/232. Complex number z satisfies $\left| \frac{3z+2i}{z+2i} \right| \leq \sqrt{3}$, where $i = \sqrt{-1}$. Least positive argument of $z + \frac{4}{\sqrt{3}}i$ will be of the form $\frac{\pi}{k}$, where k is equal to _____.

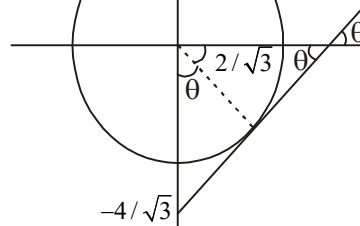
सम्मिश्र संख्या z , $\left| \frac{3z+2i}{z+2i} \right| \leq \sqrt{3}$ को संतुष्ट करती है। $z + \frac{4}{\sqrt{3}}i$ का न्यूनतम धनात्मक कोणांक $\frac{\pi}{k}$ रूप का है, जहाँ k का मान _____ होगा। (जहाँ $i = \sqrt{-1}$ है)

Ans. 3

Sol. $\left| \frac{3z+2i}{z+2i} \right| \leq \sqrt{3}$ represents the points on or inside the circle with center at origin and radius $\frac{2}{\sqrt{3}}$.

$$\cos \theta = \frac{2/\sqrt{3}}{4/\sqrt{3}} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$



A07/4/233. Let $P(z) = z^3 + az^2 + bz + c$ where $a, b, c \in \mathbb{R}$ there exists a complex number w such that the three roots of $P(z) = 0$ are $w + 3i$, $w + 9i$ and $2w - 4$ where $i^2 = -1$. Find the unit digit of the sum $|a + b + c|$ _____.

माना कि $P(z) = z^3 + az^2 + bz + c$ जहाँ $a, b, c \in \mathbb{R}$ एक सम्मिश्र संख्या w मौजूद है जिससे $P(z) = 0$ के तीन मूल $w + 3i$, $w + 9i$ और $2w - 4$ हैं जहाँ $i^2 = -1$ है। योग $|a + b + c|$ का इकाई अंक _____ होगा।

Ans. 6

Sol. $\because a, b, c \in \mathbb{R}$

$\therefore$ Equation $z^3 + az^2 + bz + c = 0$ has a real root and other 2 roots be conjugate of each other.

(I) If $2w - 4 \in \mathbb{R} \Rightarrow w \in \mathbb{R}$

But $w + 3i$ and $w + 9i$ are not conjugate of each other (not possible).

(II) If $w + 3i \in \mathbb{R}$

Let $w + 3i = \alpha \Rightarrow w = \alpha - 3i$

$$\therefore w + 9i = \alpha + 6i$$

$$2w - 4 = (2\alpha - 4) - 6i \quad \left. \vphantom{2w - 4} \right\} \text{Conjugate of each other}$$

$$\Rightarrow 2\alpha - 4 = \alpha \Rightarrow \alpha = 4 \Rightarrow w = 4 - 3i$$

$\therefore$ Roots are $4, 4 + 6i$ and $4 - 6i$

$$\therefore \text{Equation be } (z - 4)[z - (4 + 6i)][z - (4 - 6i)] = 0$$

$$\Rightarrow z^3 - 12z^2 + 84z - 208 = 0$$

$$\Rightarrow |a + b + c| = 136$$

A07/4/234. If $z_1, z_2, z_3, z_4, z_5, z_6$ are the roots of the equation $z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$, then the value of

$$\sum_{i=1}^6 z_i^5 + \sum_{i=1}^6 z_i^{14} - \prod_{i=1}^6 z_i \text{ is equal to } \underline{\hspace{2cm}}.$$

यदि z_1, z_2, z_3, z_4, z_5 तथा z_6 समीकरण $z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0$ के मूल हों, तो $\sum_{i=1}^6 z_i^5 + \sum_{i=1}^6 z_i^{14} - \prod_{i=1}^6 z_i$ का मान _____ होगा।

Ans. 4

Sol. Given equation

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = 0 \dots (i)$$

$$\therefore \text{Product of roots} = \prod_{i=1}^6 z_i = 1$$

Now, multiply equation (1) by $(z - 1)$, we get

$$z^7 - 1 = 0 \Rightarrow z^7 = 1 \dots (2) \text{ i.e. } 7^{\text{th}} \text{ roots of unity.}$$

$\therefore$ roots of equation (1) are the roots of equation (2) except $z = 1$.

$$\therefore \sum_{i=1}^6 z_i^5 = -1 \text{ and } \sum_{i=1}^6 z_i^{14} = 6$$

$$\therefore \text{required sum} = -1 + 6 - 1 = 4$$

A07/4/235. If r is the radius of the circle which touches the line $iz + \bar{z} + 1 + i = 0$ and has the lines $(1 - i)z = (1 + i)\bar{z}$ and $(1 + i)z - (1 - i)\bar{z} = 4i$ as its normals then find the value $4r^2$ _____.

यदि r त्रिज्या का वृत्त जो रेखा $iz + \bar{z} + 1 + i = 0$ को स्पर्श करता है और $(1 - i)z = (1 + i)\bar{z}$ और $(1 + i)z - (1 - i)\bar{z} = 4i$ जिसके अभिलम्ब हैं। तो $4r^2$ का मान _____ होगा।

Ans. 2

Sol. Clearly point of intersection of normals would be the centre of the required circle.

$$(1-i)z = (1+i)\bar{z} \Rightarrow \bar{z} = \frac{1-i}{1+i}z$$

$$(1+i)z + (1-i)\bar{z} = 4i \Rightarrow \bar{z} = \frac{4i - (1+i)z}{i-1}$$

$$\Rightarrow (1-i)(i-1)z = 4i(1+i) - (1+i)^2z$$

$$\Rightarrow z(4i) = 4i(1+i) \Rightarrow z = (1+i)$$

Now equation of tangent can be rewritten as,

$$\frac{1}{1+i}(iz + \bar{z}) + 1 = 0 \Rightarrow i(1-i)z + (1-i)\bar{z} + 2 = 0 \Rightarrow (i+1)z + (1-i)\bar{z} + 2 = 0$$

$$\text{Now distance of } z = (1+i) \text{ from this line} = \frac{|(1+i)(1+i) + (1-i)(1-i) + 2|}{2|1-i|} = \frac{1}{\sqrt{2}}$$

$$\text{Thus the equation of required circle is, } |z - (1+i)| = \frac{1}{\sqrt{2}}$$

A07/4/236. The number of integral values of 'a' for which at least one complex number z satisfies both the equality $|z - ai| = a + 4$ and the inequality $|z - 2| < 1$ _____.

'a' के पूर्णांक मानों की संख्या, जिनके लिए कम से कम एक सम्मिश्र संख्या z समिका $|z - ai| = a + 4$ और असमिका $|z - 2| < 1$ दोनों को संतुष्ट करती है, होगी _____

Ans. 2

Sol. To check touching conditions

$$C_1 = (0, a), C_2 = (2, 0)$$

$$r_1 = a + 4, r_2 = 1$$

$$|r_1 - r_2| \leq C_1 C_2 \leq r_1 + r_2$$

$$|a + 3| \leq \sqrt{a^2 + 4} \leq a + 5$$

$$(a + 3)^2 \leq a^2 + 4 \leq (a + 5)^2$$

$$6a + 5 \leq 0 \text{ \& } 10a + 21 \geq 0$$

$$a \leq -\frac{5}{6} \text{ \& } a \geq -\frac{21}{10}$$

$$a \in \left[-\frac{21}{10}, -\frac{5}{6}\right]$$

no. of possibilities are 2

A07/4/237. Consider lines $z(1 - mi) - \bar{z}(1 + mi) = 0$ and $z(1 - ni) - \bar{z}(1 + ni) = 0$. Let $\arg(z')$ be the angle between

them such that $\arg(z') \in \left[0, \frac{\pi}{2}\right]$. There is another line $z(1+i) - \bar{z}(-1+i) + 2i = 0$ which intersects the

previous lines in A and B such that $AB - OA = OB - AB$. Then, $\text{Max}(\arg(z'))$ is $\frac{\pi}{k}$, where k is _____.

माना रेखाएँ $z(1 - mi) - \bar{z}(1 + mi) = 0$ तथा $z(1 - ni) - \bar{z}(1 + ni) = 0$ है। माना इनके मध्य कोण $\arg(z')$ इस प्रकार है कि $\arg(z') \in \left[0, \frac{\pi}{2}\right]$ है। एक अन्य रेखा $z(1+i) - \bar{z}(-1+i) + 2i = 0$ है जो कि पहले वाली रेखाओं को बिन्दु A तथा

B पर इस प्रकार काटती है कि $AB - OA = OB - AB$ है तो $\text{Max}(\arg(z'))$ का मान $\frac{\pi}{k}$ है जहाँ k का मान _____ होगा।

Ans.

3

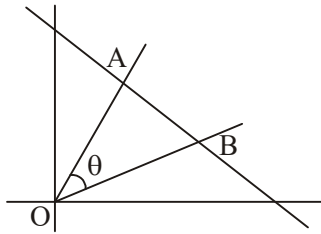
Sol.

$$\theta = \arg(z')$$

$$\cos \theta = \frac{OA^2 + OB^2 - AB^2}{2.OA.OB}$$

$$\cos \theta = \frac{(OA)^2 + (OB)^2 - \frac{1}{4}(OA+OB)^2}{2.OA.OB} = \frac{8\left(\frac{OA}{OB} + \frac{OB}{OA}\right) - \frac{1}{4}}{3}$$

$$\text{For } \theta \text{ (Maximum)} \cos \theta = \frac{1}{2}$$



A07/4/238. There exists a polynomial P of degree 5 with the property that if z is a complex number such that

$z^5 + 2004z = 1$ then $P(z^2) = 0$. Find the value of $\left\lfloor \frac{P(1)}{P(-1)} \right\rfloor$ (where $\lfloor . \rfloor$ denote greatest integer function)

माना P एक 5 घात का बहुपदीय फलन इस प्रकार विद्यमान है जिसका गुणधर्म है कि यदि z एक सम्मिश्र संख्या इस प्रकार है कि

$z^5 + 2004z = 1$ तो $P(z^2) = 0$. $\left\lfloor \frac{P(1)}{P(-1)} \right\rfloor$ का मान _____ होगा। (जहाँ $\lfloor . \rfloor$ महत्तम पूर्णांक फलन को निरूपित करता है।)

Ans.

0

Sol. $\because P(z^2) = 0$ and $z^5 + 2004z = 1$
 Let $z^2 = t \Rightarrow P(t) = 0$ and $z(z^4 + 2004) = 1$
 $\Rightarrow p(t) = 0$ and $(z(t^2 + 2004))^2 = 1$
 $\Rightarrow p(t) = 0$ and $t(t^2 + 2004)^2 - 1 = 0$
 $\Rightarrow p(t) = t(t^2 + 2004)^2 - 1$
 $\therefore \left| \frac{P(1)}{P(-1)} \right| = \left| \frac{2005^2 - 1}{2005^2 + 1} \right| \Rightarrow \left[\left| \frac{P(1)}{P(-1)} \right| \right] = 0$

A07/4/239. Consider the curves $z^2 - \bar{z}^2 = 4ic$, ($c \in \mathbb{R}$) and $z\bar{z} = 1$. If they touch at exactly two points A, B, then $|AB|$ is _____.

माना वक्र $z^2 - \bar{z}^2 = 4ic$, ($c \in \mathbb{R}$) तथा $z\bar{z} = 1$ है। यदि यह वक्र ठीक दो बिन्दु A, B पर स्पर्श करते हैं, तो $|AB|$ का मान _____ होगा।

Ans. 2

Sol. Substituting $y = \frac{c}{x}$ in $x^2 + y^2 = 1$, we get $x^4 - x^2 + c^2 = 0$ (which has equal roots)

$\Rightarrow c^2 = \frac{1}{4}$ and point can be $\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right)$ and $\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)$
 $\therefore |AB| = 2$

A07/4/240. If the area of the polygon whose vertices are the solutions (in the complex plane) of the equation $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$, can be expressed in the simplest form as $\frac{a\sqrt{b} + c}{d}$, find the value of $(a + b + c + d)$ _____.

यदि समीकरण $x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 = 0$ के हलों को सम्मिश्र तल में निरूपित करने पर बने बहुभुज का क्षेत्रफल सरलतम रूप में $\frac{a\sqrt{b} + c}{d}$ से व्यक्त किया जा सकता है, तो $(a + b + c + d)$ का मान _____ होगा।

Ans. 8

Sol. Centre $\circ \frac{k^2\beta - \alpha}{k^2 - 1}$, Radius $= \frac{1}{(k^2 - 1)} \sqrt{|\alpha - k^2\beta|^2 - (k^2 \cdot |\beta|^2 - |\alpha|^2) \cdot (k^2 - 1)}$

A07/4/241. If $z_1, z_2, z_3 \in \mathbb{C}$ satisfying $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ and $z_1 z_2 z_3 = 1$. Also $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$. Then find the value of $[|z_1 + z_2 + z_3|]$, where $[.]$ denotes the greatest integer function _____.

यदि $z_1, z_2, z_3 \in \mathbb{C}$, $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ तथा $z_1 z_2 z_3 = 1$ को संतुष्ट करता है तथा $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$, तब $[|z_1 + z_2 + z_3|]$ का मान ज्ञात कीजिए, जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है।

Ans. 2

Sol. $z_1 z_2 + z_2 z_3 + z_3 z_1$

$$= z_1 z_2 z_3 (\bar{z}_1 + \bar{z}_2 + \bar{z}_3)$$

$$\Rightarrow z_1 z_2 + z_2 z_3 + z_3 z_1 = 1$$

$\therefore z_1, z_2, z_3$ satisfy संतुष्ट होता है

$$z^3 - z^2 + z - 1 = 0$$

or या $z_1 = -i$

$$z_2 = 1$$

$$z_3 = i \Rightarrow |z_1 + z_2^2 + z_3^3| = |-i + 1 - i| = \sqrt{5}$$

A07/4/242. Let $z_1, z_2 \in \mathbb{C}$ such that $|z_1 - 3i| + |z_1 + 4| = 5$ and $|z_2 - 3i| + |z_2 + 4| = 8$. If maximum value of $|z_1 - z_2|$ is $\frac{p}{q}$ where p, q are relatively prime numbers then find the value of $(p - 2q)$. _____.

माना $z_1, z_2 \in \mathbb{C}$ इस प्रकार है कि $|z_1 - 3i| + |z_1 + 4| = 5$ और $|z_2 - 3i| + |z_2 + 4| = 8$ यदि $|z_1 - z_2|$ का अधिकतम मान $\frac{p}{q}$ है जहाँ p, q सह अभाज्य संख्याएँ हैं तब $(p - 2q)$ का मान _____ होगा।

Ans.

9

Sol.

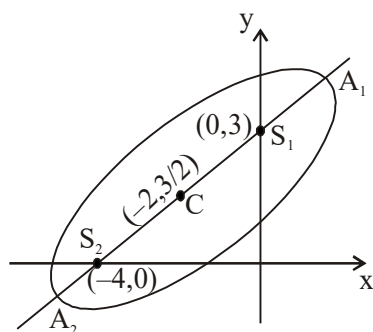
$$|z_1 - 3i| + |z_1 + 4| = 5$$

$\Rightarrow z_1$ lies on the line segment joining $(0, 3)$ and $(-4, 0)$.

$$|z_2 - 3i| + |z_2 + 4| = 8$$

$\Rightarrow z_2$ lies on the ellipse whose foci are $(0, 3)$ and $(-4, 0)$ and whose length of major axis is 8.

From the figure it is clear that



$$|z_1 - z_2|_{\max} = A_1 S_2 = CS_2 + CA_1$$

$$= \frac{5}{4} + 4 = \frac{12}{2}$$

$$\therefore p - 2q = 12 - 2 \times 2 = 8$$

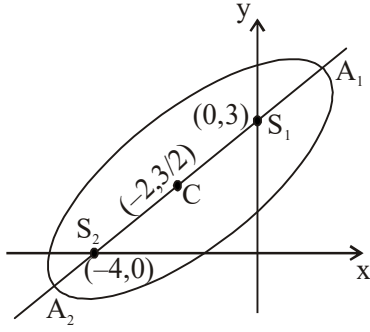
$$|z_1 - 3i| + |z_1 + 4| = 5$$

$\Rightarrow (0, 3)$ और $(-4, 0)$ को मिलाने वाली रेखा z_1 स्थित है।

$$|z_2 - 3i| + |z_2 + 4| = 8$$

$\Rightarrow z_2$ दीर्घवृत्त पर स्थित है जिसकी नाभि $(0, 3)$ तथा $(-4, 0)$ और जिसकी दीर्घ अक्ष की लम्बाई 8 है।

चित्र से यह स्पष्ट है कि



$$|z_1 - z_2|_{\max} = A_1 S_2 = CS_2 + CA_1$$

$$= \frac{5}{4} + 4 = \frac{12}{2}$$

$$\therefore p - 2q = 13 - 2 \times 2 = 9$$

A07/4/243. Let n be a number amongst the first hundred natural numbers such that $(\sin\theta + i\cos\theta)^n = \sin(n\theta) + i\cos(n\theta)$, then find number of possible values of n _____.

माना n एक संख्या है जो कि प्रथम सौ प्रकृत संख्याओं में से ली गई है जबकि $(\sin\theta + i\cos\theta)^n = \sin(n\theta) + i\cos(n\theta)$, तब n के संभावित मानों की संख्या _____ होगी।

Ans. 25

Sol. $\left\{ \cos\left(\frac{\pi}{2} - \theta\right) + i\sin\left(\frac{\pi}{2} - \theta\right) \right\}^n = \cos\left(\frac{n\pi}{2} - n\theta\right) + i\sin\left(\frac{n\pi}{2} - n\theta\right)$ (DeMoivre's Theorem द:मयावर प्रमेय)

$$= \sin(n\theta) + i\cos(n\theta)$$

Hence अतः $n = 4k + 1 \Rightarrow n = 1, 5, 9, \dots, 97$

$\Rightarrow$ Number of value is मानों की संख्या 25 है।

A07/4/244. If $A = \left\{ z : \left| \frac{z-2}{z+2} \right| = 3, z \in \mathbb{C} \right\}$ and $z_1, z_2, z_3, z_4 \in A$ are 4 complex number representing points P, Q, R, S respectively on the complex plane such that $z_1 - z_2 = z_4 - z_3$, then maximum value of area of quadrilateral PQRS is _____.

यदि $A = \left\{ z : \left| \frac{z-2}{z+2} \right| = 3, z \in \mathbb{C} \right\}$ और $z_1, z_2, z_3, z_4 \in A$ चार सम्मिश्र संख्याएँ हैं जो सम्मिश्र तल पर क्रमशः बिन्दु P, Q, R, S को इस प्रकार निरूपित करती हैं कि $z_1 - z_2 = z_4 - z_3$ हो, तो चतुर्भुज PQRS के क्षेत्रफल का अधिकतम मान _____ होगा।

Ans. 4.5

Sol. Let $z = x + iy$

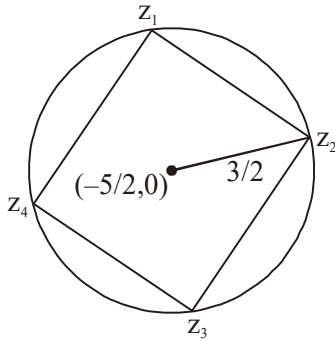
$$\Rightarrow |x - 2 + iy| = 3|x + iy + 2|$$

$$\begin{aligned}\Rightarrow (x-2)^2 + y^2 &= 9(x+2)^2 + 9y^2 \\ \Rightarrow 8x^2 + 8y^2 + 40x + 32 &= 0 \\ \Rightarrow x^2 + y^2 + 5x + 4 &= 0\end{aligned}$$

Clearly locus is a circle with centre $\left(-\frac{5}{2}, 0\right)$ and radius $= \sqrt{\frac{25}{4} - 4} = \frac{3}{2}$

since $z_1, z_2, z_3, z_4 \in A$

$\Rightarrow$ all these points must lie on this circle and forms a parallelogram, since $z_1 + z_3 = z_2 + z_4$



Now to have maximum area of parallelogram, it must be a square whose area will be

$$\frac{d^2}{2} = \frac{9}{2}$$

A07/4/245. If $\alpha = e^{\frac{2\pi i}{7}}$ and $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$, such that $\sum_{r=0}^6 f(\alpha^r x) = n(A_0 + A_n x^n + A_{2n} x^{2n})$, then the value of n is _____.

यदि $\alpha = e^{\frac{2\pi i}{7}}$ तथा $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$, इस प्रकार है कि $\sum_{r=0}^6 f(\alpha^r x) = n(A_0 + A_n x^n + A_{2n} x^{2n})$, तो n का मान _____ होगा।

Ans. 7

Sol. $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k = \sum_{k=0}^{20} A_k x^k$

$$\begin{aligned}\therefore \sum_{r=0}^6 f(\alpha^r x) &= f(x) + f(\alpha x) + f(\alpha^2 x) + f(\alpha^3 x) + f(\alpha^4 x) + f(\alpha^5 x) + f(\alpha^6 x) \\ &= \sum_{k=0}^{20} \left(A_k x^k + A_k (\alpha x)^k + A_k (\alpha^2 x)^k + \dots + A_k (\alpha^6 x)^k \right) \\ &= \sum_{k=0}^{20} \left\{ A_k x^k \left(1 + \alpha^k + (\alpha^2)^k + (\alpha^3)^k + \dots + (\alpha^6)^k \right) \right\}\end{aligned}$$

$$= A_0 x^0 (7) + A_7 x^7 (7) + A_{14} x^{14} (7)$$

$$= 7[A_0 + A_7 x^7 + A_{14} x^{14}] \Rightarrow n = 7$$

A08/4/246. An urn contains 3 white and 7 black balls. Balls are drawn one by one from urn. Each time a ball is drawn, it is replaced by a black one. If probability of drawing out all white balls in the 4th draw is p , then $25\sqrt{p}$ is equal to _____.

एक कलश में 3 सफेद और 7 काली गेंदें हैं। कलश से एक-एक करके गेंदे निकाली जाती हैं। हर बार एक गेंद निकालने पर, उसे एक काली गेंद से बदल दिया जाता है। यदि चौथी बार सभी सफेद गेंदों के निकालने की संभावना p है, तो $25\sqrt{p}$ के बराबर _____ होगा।

Ans. 3

Sol. Case-I : I II III IV

$$W \ W \ B \ W \rightarrow \frac{3}{10} \times \frac{2}{10} \times \frac{9}{10} \times \frac{1}{10}$$

Case-I : I II III IV

$$W \ B \ W \ W \rightarrow \frac{3}{10} \times \frac{8}{10} \times \frac{2}{10} \times \frac{1}{10}$$

Case-I : I II III IV

$$B \ W \ W \ W \rightarrow \frac{7}{10} \times \frac{3}{10} \times \frac{2}{10} \times \frac{1}{10}$$

$$\text{Required probability } p = \frac{144}{10^4}$$

A08/4/247. The number of possible integral values of k such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y + k^2 = 0$ have exactly two common tangents is λ . A biased coin which comes up heads three times as often as tails is tossed. If it shows heads, a chip is drawn from urn-I which contains 2 white chips and 5 red chips. If the coin comes up tails, a chip is drawn from urn-II which contains λ white and 4 red chips. Given that a red chip was drawn, what is the probability that the coin came up heads?

एक सिक्के को उछालने पर चित आने की प्रायिकता पट से 3 गुना है। यदि यह चित दिखाता है तो बक्से -I में से एक चिप निकाली जाती है जिसमें दो सफेद तथा 5 लाल चिप हैं। यदि पट आता है तो बक्से -II में से एक चिप निकाली जाती है जिसमें λ सफेद तथा 4 लाल चिप हैं। दिया गया है कि निकाली गई चिप लाल है, तो प्रायिकता ज्ञात कीजिये कि सिक्के को उछालने पर चित आया है _____.

Ans. 165/193

Sol.

A08/4/248. A pair of dice is rolled together till a sum of either 5 or 7 is obtained, then the probability that 5 comes before

7 is m/n . Where $\gcd(m, n) = 1$. If α, β, γ are the roots of $x^3 + mx^2 + n = 0$, then the value of $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ is

equals to _____.

पासों के एक युग्म को तब तक एक साथ घुमाया जाता है जब तक कि 5 या 7 का योग प्राप्त न हो जाए, तब 7 से पहले 5 आने

की प्रायिकता m/n है। जहाँ $\gcd(m, n) = 1$ है। यदि $x^3 + mx^2 + n = 0$ के मूल α, β, γ हैं, तो $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ का मान

_____ बराबर होगा।

Ans. 8

Sol. P(5 before 7)

$$= \frac{1}{9} + \frac{13}{18} \times \frac{1}{9} + \left(\frac{13}{18}\right)^2 \times \frac{1}{9} + \dots \infty$$

A08/4/249. There are 10 cards, labeled from 1 to 10. Three cards denoted by a, b, c ($a > b > c$) are drawn from the cards at the same time if the probability such that $\int_0^a (x^2 - 2bx + 3c)dx = 0$ is $\frac{1}{k}$ then $\frac{k}{10}$ is _____.

1 से 10 तक नामांकित 10 पत्ते हैं। तीन पत्तों को क्रमशः a, b, c ($a > b > c$) से व्यक्त किया जाता है, एक साथ पत्तों से खींचा जाता है यदि प्रायिकता $\frac{1}{k}$ है जबकि $\int_0^a (x^2 - 2bx + 3c)dx = 0$ तब $\frac{k}{10}$ का मान _____ होगा।

Ans. 4

Sol. $\int_0^a (x^2 - 2bx + 3c)dx = 0$

$$\Rightarrow \frac{a^3}{3} - ba^2 + 3ac = 0$$

$$\Rightarrow a^2 = 3ab + 9c = 0$$

$$\Rightarrow a^2 = 3(ab - 3c)$$

$$\Rightarrow a \text{ is multiple of } 3$$

$$\text{Case-I: } a = 3 \Rightarrow b = 2, c = 1$$

$$\text{Case-II: } a = 6 \Rightarrow b = 3, c = 2$$

$$\text{Case-III: } a = 9 \Rightarrow b = 4, c = 3$$

$$\Rightarrow \text{only 3 solution of } (a, b, c)$$

$$\Rightarrow \text{Probability equal to } \frac{3}{{}^{10}C_3} = \frac{1}{40}$$

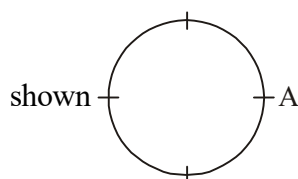
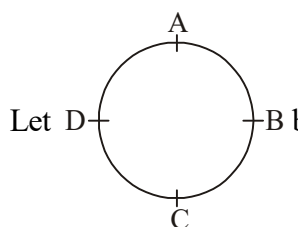
A08/4/250. Four people sit around a round table, each person knowing his two adjacent only the four get up, walk around, and sit back down at the random seats, the probability that someone is sitting next to some-one he doesn't know is $\frac{p}{q}$, where p and q are relatively prime natural numbers, then find the value of $p+q$ _____.

चार व्यक्ति एक गोल मेज के चारों ओर बैठे हैं। प्रत्येक व्यक्ति अपने आसपास वालों को ही जानता है। चारों व्यक्ति अपने स्थानों से उठकर, घुमते हैं तथा यदृच्छया, सीटों पर बैठ जाते हैं। यदि कोई व्यक्ति, अन्य व्यक्ति के पास बैठा है तथा उसे नहीं जानता

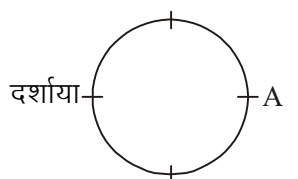
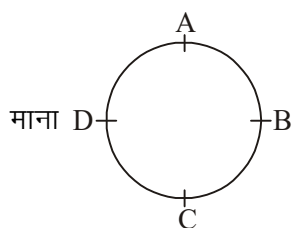
है कि प्रायिकता $\frac{p}{q}$ जहाँ p और q सहअभाज्य हैं तब $p+q$ का मान _____ होगा।

Ans. 5

Sol. Let D— B be the original arrangement after walking around, WLOG, let A, sit be chosen, as



Now if c sits adjacent to him there is a pair of neighbors not knowing each other, so required probability is $\frac{2}{3}$.



अब यदि c उसके आसपास बैठता है तथा उनको, आपस में नहीं जानता है इसलिए अभीष्ट प्रायिकता $\frac{2}{3}$ है।

A08/4/251. From a pack of 52 playing cards, half of the cards are randomly removed without looking at them. From the remaining cards, 3 cards are drawn randomly. The probability that all are king, is $\frac{1}{\lambda}$, then find the number of prime divisors of λ . _____.

52 ताश की गड्डी से आधे पत्ते को यादृच्छिक बिना देखे हुए हटाया जाता है शेष पत्तों से 3 पत्ते यादृच्छिक निकाले जाते हैं तब सभी के राजा होने की प्रायिकता $\frac{1}{\lambda}$ है तब λ के अभाज्य भाजकों की संख्या _____ होगी।

Ans. 4

Sol.

$$P(A|E) = \frac{P(A \cap E)}{P(E)} = \frac{\sum_{i=0}^4 P(E_i)P(A|E_i)}{\sum_{i=0}^4 P(E_i)} = \frac{\frac{{}^{48}C_{26}}{{}^{52}C_{26}} \times \frac{{}^4C_3}{{}^{26}C_3} + \frac{{}^4C_1 {}^{48}C_{25}}{{}^{52}C_{26}} \times \frac{{}^3C_3}{{}^{26}C_3} + 0 + 0 + 0}{\frac{{}^4C_0 {}^{48}C_{26} + {}^4C_1 {}^{48}C_{25} + {}^4C_2 {}^{48}C_{24} + {}^4C_3 {}^{48}C_{23} + {}^4C_4 {}^{48}C_{22}}{{}^{52}C_{26}}}$$

$$= \frac{4({}^{48}C_{26} + {}^{48}C_{25})}{{}^{26}C_3 ({}^{52}C_{26})} = \frac{4(49!)3!23!26!26!}{26!23!26!52!} = \frac{4(3!)}{52 \times 51 \times 50}$$

$$= \frac{1}{13 \cdot 17 \cdot 25}$$

A08/4/252. The probability for which $\lim_{x \rightarrow 0} \frac{\ell n((\cos x)^a)}{x^b}$ exists finitely, where (a, b) is the possible outcome of throwing a pair of normal dice, is $\frac{p}{q}$ (where p and q are coprime) then find the value of (q - p) _____.

प्रायिकता $\frac{p}{q}$ होगी जबकि $\lim_{x \rightarrow 0} \frac{\ell n((\cos x)^a)}{x^b}$ परिमित रूप से विद्यमान है जहाँ (a, b) साधारण पासे युग्म को फेंकने पर संभावित परिणाम है। जहाँ $\frac{p}{q}$ (जहाँ p और q अभाज्य संख्याएँ हैं) तब (q - p) का मान _____ होगा।

Ans. 2

Sol. $\lim_{x \rightarrow 0} \frac{a \ell n(\cos x)}{x^b}$ exists finitely

$$\lim_{x \rightarrow 0} \frac{a(\cos x - 1)}{x^b} = \frac{-ax^2}{ab^2} \Rightarrow b \leq 2 \Rightarrow b = 1, 2$$

$$\text{Probability} = \frac{2}{6} = \frac{1}{3}$$

A08/4/253. All the face cards from a pack of 52 playing cards are removed. From the remaining pack half of the cards are randomly removed without looking at them and then randomly drawn two cards simultaneously from the

remaining. If the probability that two cards drawn are both aces is $\frac{p \binom{38}{20}}{40 \binom{20}{2}}$, find p. _____.

52 पत्तों की तास की गड्डी से सभी चेहरे रखने वाले पत्तों को हटाया जाता है शेष गड्डी में से पत्तों का बिना देखे हुये यादृच्छिक आधे पत्तों को हटया जाता है तब शेष पत्तों से यादृच्छिक दो पत्ते एक साथ खिंचे जाते हैं। यदि खिंचे गये दो पत्तों के दोनों इक्के

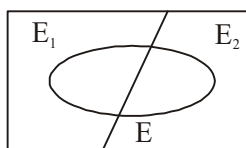
होने की प्रायिकता $\frac{p \binom{38}{20}}{40 \binom{20}{2}}$ तब p का मान _____ होगा।

Ans. 6

Sol. 52 $\xrightarrow{\text{face card removed}}$ 40 $\xrightarrow{20 \text{ drawn randomly}}$

Let E_0 : 20 cards randomly removed has no aces.
 E_1 : 20 cards randomly removed has exactly one ace.
 E_2 : 20 cards randomly removed has exactly 2 aces.
 E : event that 2 drawn from the remaining 20 cards has both the aces.

$$P(E) = P(E \cap E_0) + P(E \cap E_1) + P(E \cap E_2) \\ = P(E_0) \cdot P(E / E_0) + P(E_1) \cdot P(E / E_1) + P(E_2) \cdot P(E / E_2)$$



= 40 $\begin{cases} 4 \text{ aces} \\ 36 \text{ other} \end{cases}$

$$= \frac{{}^4C_0 \cdot {}^{36}C_{20}}{40 \binom{20}{2}} + \frac{{}^4C_1 \cdot {}^{36}C_{19}}{40 \binom{20}{2}} + \frac{{}^4C_2 \cdot {}^{36}C_{18}}{40 \binom{20}{2}} + \frac{{}^2C_2}{{}^{20}C_2}$$

$$= \frac{{}^{36}C_{20} \cdot {}^4C_2 + {}^4C_1 \cdot {}^{36}C_{19} \cdot {}^3C_2 + {}^4C_2 \cdot {}^{36}C_{18} \cdot {}^2C_2}{40 \binom{20}{2}}$$

$$= \frac{6 \cdot {}^{36}C_{20} + 12 \cdot {}^{36}C_{19} + 6 \cdot {}^{36}C_{18}}{40 \binom{20}{2}}$$

$$= \frac{6 \left[{}^{36}C_{20} + {}^{36}C_{19} + {}^{36}C_{19} + {}^{36}C_{18} \right]}{40 \binom{20}{2}}$$

$$= \frac{6 \left({}^{37}C_{20} + {}^{37}C_{19} \right)}{40 \binom{20}{2}} = \frac{6 \binom{38}{20}}{40 \binom{20}{2}}$$

$$\Rightarrow p = 6$$

A08/4/254. Humpty plays three matches against Dumpty. In any match the probabilities of Humpty getting 0, 2 and 3 points are 0.2, 0.3 and 0.5 respectively. If probability that Humpty getting atmost 6 points is $\frac{p}{200}$, then then sum of the digits of 'p' is _____.

हम्प्टी एवं डम्प्टी साथ में तीन मैच खेलते हैं। किसी मैच में हम्प्टी को 0, 2 तथा 3 अंक मिलने की प्रायिकताएँ क्रमशः 0.2, 0.3 तथा 0.5 हैं। यदि हम्प्टी को अधिकतम 6 अंक मिलने की प्रायिकता $\frac{p}{200}$ है, तो 'p' के अंकों का योग होगा _____.

Ans. 4

Sol. Possible points that Humpty Not scores are = $\begin{cases} 9 & \text{i.e. } 333 \\ 8 & \text{i.e. } 233 \\ 7 & \text{i.e. } 223 \end{cases}$

$$\begin{aligned} \text{Required probability} &= 1 - P(\text{he score 7, 8 or 9}) \\ &= 1 - \{ {}^3C_2(0.3)^2 \cdot (0.5) + {}^3C_2 \cdot (0.5)^2 \cdot (0.3) + {}^3C_3 \cdot (0.5)^3 \} \\ &= 1 - \left\{ \frac{3 \cdot 9 \cdot 5 + 3 \cdot 25 \cdot 3 + 1 \cdot 125}{1000} \right\} = \frac{103}{200} \end{aligned}$$

MATCH THE COLUMN TYPE QUESTION(S)

A01/5/255. A Polynomial $p(x) = x^6 + 3x^5 - 3x^4 + ax^3 + bx^2 + cx + d$, is given such that all roots of $p(x) = 0$ are equal to either α or β ($\alpha, \beta \in I$) then match the following :

बहुपद $p(x) = x^6 + 3x^5 - 3x^4 + ax^3 + bx^2 + cx + d$ इस प्रकार है कि $p(x) = 0$ के सभी मूल बराबर हैं जो α या β के बराबर हैं ($\alpha, \beta \in I$) तब सुमेलित कीजिए :

Column-I

- (a) $a =$
- (b) $b =$
- (c) $c =$
- (d) $d =$

Column-II

- (p) -8
- (q) 12
- (r) 6
- (s) -11

Choose the correct option :

सही विकल्प का चयन कीजिए :

- (A) (a) $\rightarrow$ (p), (b) $\rightarrow$ (q), (c) $\rightarrow$ (r), (d) $\rightarrow$ (s) (B) (a) $\rightarrow$ (q), (b) $\rightarrow$ (r), (c) $\rightarrow$ (s), (d) $\rightarrow$ (p)
 (C) (a) $\rightarrow$ (r), (b) $\rightarrow$ (s), (c) $\rightarrow$ (p), (q) $\rightarrow$ (p) (D) (a) $\rightarrow$ (s), (b) $\rightarrow$ (r), (c) $\rightarrow$ (q), (d) $\rightarrow$ (p)

Ans. D

Sol. $p(x) = x^6 + 3x^5 - 3x^4 + ax^3 + bx^2 + cx + d$

Sum of roots $= -3$ _____ (1)

Sum of roots taken two at a time $= -3$ _____ (2)

Given, roots are integers [either m or n]

- C-1 If all roots are α i.e., $\alpha, \alpha, \alpha, \alpha, \alpha, \alpha, \alpha$ Then sum of roots $= 6\alpha = -3$ (from (i)) Not possible as L.H.S. is multiple of '6' but R.H.S. is not.
- C-2 If 5 roots are α and 1 is β
i.e., $\alpha, \alpha, \alpha, \alpha, \alpha, \alpha$
Sum $= 5\alpha + \beta = -3$
sum of roots taken two at a time $= 10\alpha^2 + 5\alpha\beta = -3$
Not possible as L.H.S. is multiple of '5' but R.H.S. is not.
- C-3 If 4-roots are ' α ' and 2 are ' β ' i.e., $\alpha, \alpha, \alpha, \alpha, \alpha, \beta, \beta \Rightarrow$ Sum $= 4\alpha + 2\beta = -3$
Not possible as L.H.S. is even and R.H.S. is odd.
- C-4 If roots are $\alpha, \alpha, \alpha, \beta, \beta, \beta$
Then sum $= 3\alpha + 3\beta = -3 \Rightarrow \alpha + \beta = -1$ _____(3)
Sum taken two at a time $= 3\alpha^2 + 3\beta^2 + 9\alpha\beta = -3$
 $\Rightarrow \alpha^2 + \beta^2 + 3\alpha\beta = -1$ _____(4)
(4) - (3)² $\Rightarrow \alpha\beta = -2$ _____(5)
 $\Rightarrow \alpha = -2$ and $\beta = 1$ OR $\alpha = 2$ and $\beta = -1$ _____(6)
From (3) and (6) $\Rightarrow \alpha = -2$ & $\beta = 1$
i.e. roots are $-2, -2, -2, 1, 1, 1$. Now a = -(sum of roots taken '3' at a time)
 $= -(-8 + 3 \cdot 3 \cdot 2^2 \cdot 1 + 3 \cdot 3 \cdot (-2) \cdot 1^2 + 1^3)$
 $= -(-8 + 36 + 1 - 18) = -11$
b = sum of roots taken '4' at a time
 $= 3 \cdot (-2)^3 \cdot (1) + 3 \cdot (-2)^2 \cdot 1^2 \cdot 3 + 3 \cdot (-2) \cdot 1^3$
 $= 24 + 36 - 6 = 6$
c = -(sum of roots taken '5' at a time) $= (3 \cdot (-2)^3 \cdot 1^2 + 3 \cdot (-2)^2 \cdot (1)^3)$
 $= -(-24 + 12) = 12$
d = product of roots $= (-2)^3 (1)^3 = -8$.

A01/5/256. Match the following column :

- | Column-I | Column-II |
|--|-----------|
| (a) Number of real solution of $ x + 1 = e^x$ is | (p) 2 |
| (b) The number of non-negative real roots of $2^x - x - 1 = 0$ equal to | (q) 3 |
| (c) If p and q be the roots of the quadratic equation $x^2 - (\alpha - 2)x - \alpha - 1 = 0$, then minimum value of $p^2 + q^2$ is equal to | (r) 6 |
| (d) If α and β are the roots of $2x^2 + 7x + c = 0$ and $ \alpha^2 - \beta^2 = \frac{7}{4}$, then c is equal to | (s) 4 |
| | (t) 5 |

Choose the correct option :

निम्नलिखित स्तम्भों को सुमेलित कीजिए :

- (a) $|x + 1| = e^x$ के हलों की संख्या होगी (p) 2
 (b) $2^x - x - 1 = 0$ के अऋणात्मक वास्तविक मूलों की संख्या होगी (q) 3
 (c) यदि p तथा q द्विघात समीकरण $x^2 - (\alpha - 2)x - \alpha - 1 = 0$ के मूल हैं, तो $p^2 + q^2$ का न्यूनतम मान होगा (r) 6
 (d) यदि α तथा β समीकरण $2x^2 + 7x + c = 0$ के मूल हैं तथा $|\alpha^2 - \beta^2| = \frac{7}{4}$ है, तो c बराबर होगा (s) 4

(t) 5

सही विकल्प का चयन कीजिए :

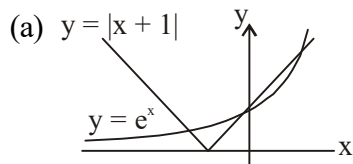
- (A) $a \rightarrow q; b \rightarrow p; c \rightarrow t; d \rightarrow s$
 (C) $a \rightarrow q; b \rightarrow p; c \rightarrow t; d \rightarrow r$

- (B) $a \rightarrow p; b \rightarrow q; c \rightarrow t; d \rightarrow r$
 (D) $a \rightarrow q; b \rightarrow t; c \rightarrow p; d \rightarrow r$

Ans.

C

Sol.



Number of solution is 3

(b) $2^x - x - 1 = 0$

Consider are two solutions

$\therefore$ there are two solutions

$x = 0, 1$ (both are non-negative)

(c) $p + q = \alpha - 2$

$pq = -\alpha - 1$

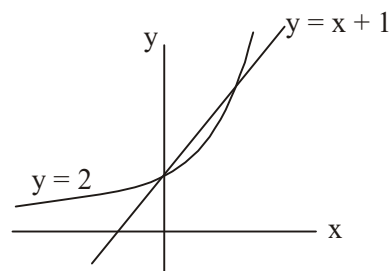
$\therefore p^2 + q^2 = (\alpha - 2)^2 - 2(-\alpha - 1) = \alpha^2 - 4\alpha + 4 + 2\alpha + 2 = \alpha^2 - 2\alpha + 6 = (\alpha - 1)^2 + 5$

$\therefore$ least value of $p^2 + q^2 = 5$

(d) $\alpha + \beta = -\frac{7}{2}, \alpha\beta = \frac{c}{2}$

$\therefore \frac{7}{4} = |\alpha^2 - \beta^2| = |(\alpha + \beta)| \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \frac{7}{2} \sqrt{49 - 8c}$

$49 - 8c = 1 \Rightarrow c = 6$



A02/5/257. Note : A sequence is called an H.P. if the reciprocal of its terms are in A.P.
 Match the following column :

Column-I

Column-II

- (a) Number of positive integral ordered pairs of (a, b) such that 6, a , b are in harmonical progression, is

(p) 6

(b) If x, y and z are positive real numbers and $x = \frac{12 - yz}{y + z}$. The maximum (q) 7

value of (xyz) equals

(c) If the roots of the equation $3x^3 - 26x^2 + 52x - 24 = 0$ are in G.P. (r) 8

then the largest root, is

(s) 9

Choose the correct option :

नोट : एक श्रेणी H.P. होती है यदि पदों के व्युत्क्रम A.P. में होते हैं

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

(a) धनात्मक क्रमित युग्मों (a, b) की संख्या यदि 6, a, b ह. श्रेणी बनाते हैं

(p) 6

(b) यदि x, y तथा z धनात्मक वास्तविक संख्याएँ हैं तथा $x = \frac{12 - yz}{y + z}$ है तो

(q) 7

(xyz) का अधिकतम मान होगा

(c) यदि समीकरण $3x^3 - 26x^2 + 52x - 24 = 0$ के मूल G.P. में हैं,

(r) 8

तो सबसे बड़ा मूल होगा

(s) 9

सही विकल्प का चयन कीजिए :

(A) $a \rightarrow q ; b \rightarrow r ; c \rightarrow p$

(B) $a \rightarrow r ; b \rightarrow p ; c \rightarrow q$

(C) $a \rightarrow p ; b \rightarrow q ; c \rightarrow r$

(D) $a \rightarrow r ; b \rightarrow q ; c \rightarrow p$

Ans. A

Sol. (a) $6, a, b$ in H.P. $\Rightarrow \frac{1}{6}, \frac{1}{a}, \frac{1}{b}$ are in A.P.

$$\Rightarrow \frac{2}{a} = \frac{1}{6} + \frac{1}{b} \Rightarrow \frac{1}{b} = \frac{2}{a} - \frac{1}{6} \Rightarrow \frac{1}{b} = \frac{12 - a}{6a} \Rightarrow b = \frac{6a}{12 - a}$$

$$a \in \{3, 4, 6, 8, 9, 10, 11\} \Rightarrow 7 \text{ Ans.}$$

(b) Given $xy + yz + zx = 12$

$$\left. \begin{array}{l} (x - y)^2 \geq 0 \\ (y - z)^2 \geq 0 \\ (z - x)^2 \geq 0 \end{array} \right\} \text{ add } \Rightarrow x^2 + y^2 + z^2 \geq xy + yz + zx$$

$$\therefore x^2 + y^2 + z^2 \geq 12$$

Now using AM, GM in x^2, y^2, z^2

$$(xyz)^{2/3} \leq \frac{x^2 + y^2 + z^2}{3}; \therefore (xyz)^{2/3} \leq \frac{12}{3}; (xyz)^{2/3} \leq 4 \Rightarrow xyz \leq 8$$

Alternatively : Consider xy, yz, zx

$$A.M. \geq G.M.$$

$$\frac{xy + yz + zx}{3} \geq (x^2 y^2 z^2)^{1/3}$$

$$4^{3/2} \geq xyz$$

$$xyz \leq 8$$

$$(c) \frac{a}{r}, a, ar. \quad \text{Hence } a^3 = \frac{24}{3} \Rightarrow a^3 = 8 \Rightarrow a = 2$$

compute the other root as 6, 2, 2/3. Hence largest root is 6 Ans.

A02/5/258. Match the following column :

Column-I

Column-II

(a) a, b, c be positive real numbers then the least value of

(p) 1

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$$

(b) $x < 0, y > 0, z > 0$ and $x + y + z = 1$ then the least value of

(q) $\frac{3}{5}$

$$\frac{x}{2-x} + \frac{y}{2-y} + \frac{z}{2-z}$$

(c) a, b, c are positive rational number such that $a + b + c = 1$ the the maximum value of $a^a b^b c^c + a^b b^c c^a + a^c b^a c^b$ is

(r) $\frac{5}{2}$

(d) If a, b, c are positive real number $a + b + c = 3$ then the maximum

(s) $\frac{3}{2}$

$$\text{possible value of } \frac{bc}{b+c} + \frac{ac}{a+c} + \frac{ab}{a+b}, a, b, c \in \mathbb{R}^+$$

Choose the correct option :

स्तम्भ-I

स्तम्भ-II

(a) a, b, c धनात्मक वास्तविक संख्याएँ हैं तो $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$

(p) 1

का न्यूनतम मान होगा

(b) $x < 0, y > 0, z > 0$ तथा $x + y + z = 1$ तो $\frac{x}{2-x} + \frac{y}{2-y} + \frac{z}{2-z}$ (q) $\frac{3}{5}$

का न्यूनतम मान होगा

(c) a, b, c धनात्मक परिमेय संख्याएँ इस प्रकार हैं कि $a + b + c = 1$ तो (r) $\frac{5}{2}$

$a^a b^b c^c + a^b b^c c^a + a^c b^a c^b$ का अधिकतम मान होगा

(d) यदि a, b, c धनात्मक वास्तविक संख्याएँ हैं तथा $a + b + c = 3$ तब (s) $\frac{3}{2}$

$\frac{bc}{b+c} + \frac{ac}{a+c} + \frac{ab}{a+b}$, $a, b, c \in \mathbb{R}^+$ का अधिकतम संभव मान होगा

सही विकल्प का चयन कीजिए :

(A) $a - s; b - q; c - p; d - r$

(B) $a - r; b - q; c - p; d - s$

(C) $a - s; b - p; c - q; d - r$

(D) $a - r; b - p; c - q; d - s$

Ans. A

Sol. (a) $\frac{\left(\frac{b+c}{a+b+c}\right)^{-1} + \left(\frac{c+a}{a+b+c}\right)^{-1} + \left(\frac{a+b}{a+b+c}\right)^{-1}}{3} \geq \left(\frac{\frac{b+c}{a+b+c} + \frac{c+a}{a+b+c} + \frac{a+b}{a+b+c}}{3}\right)^{-1}$

$$\frac{\left(\frac{b+c}{a+b+c}\right)^{-1} + \left(\frac{c+a}{a+b+c}\right)^{-1} + \left(\frac{a+b}{a+b+c}\right)^{-1}}{3} \geq \left(\frac{2}{3}\right)^{-1} = \frac{3}{2}$$

$$1 + \frac{a}{b+c} + 1 + \frac{b}{a+c} + 1 + \frac{c}{a+b} \geq \frac{9}{2}$$

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$$

(b) $\frac{(2-x)^{-1} + (2-y)^{-1} + (2-z)^{-1}}{3} \geq \left(\frac{2-x+2-y+2-z}{3}\right)^{-1}$

$$\frac{(2-x)^{-1} + (2-y)^{-1} + (2-z)^{-1}}{3} \geq \left(\frac{6-(x+y+z)}{3}\right)^{-1}$$

$$(2-x)^{-1} + (2-y)^{-1} + (2-z)^{-1} \geq \frac{9}{5}$$

$$\frac{1}{2-x} + \frac{1}{2-y} + \frac{1}{2-z} \geq \frac{9}{5}$$

$$\frac{2}{2-x} + \frac{2}{2-y} + \frac{2}{2-z} \geq \frac{18}{5}$$

$$1 + \frac{x}{2-x} + 1 + \frac{y}{2-y} + 1 + \frac{z}{2-z} \geq \frac{18}{5}$$

$$\frac{x}{2-x} + \frac{y}{2-y} + \frac{z}{2-z} \geq \frac{18}{5} - 3$$

$$\frac{x}{2-x} + \frac{y}{2-y} + \frac{z}{2-z} \geq \frac{3}{5}$$

$$(c) \frac{ca + ab + bc}{a + b + c} \geq \left(a^c b^a c^b \right)^{1/(a+b+c)} \text{ or } ac + ab + bc \leq a^c b^a c^b$$

Similarly, $ab + bc + ac \leq a^b b^c c^a$; $aa + bb + cc \geq a^a b^b c^c$

Adding together we get

$$a^b b^c c^a + a^c b^a c^b + a^a b^b c^c \leq (a + b + c)(a + b + c)$$

$$a^b b^c c^a + a^c b^a c^b + a^a b^b c^c \leq 1$$

$$(d) \frac{b+c}{2} \geq \frac{2bc}{b+c} \text{ A.M.} \geq \text{H.M.}$$

$$\frac{2bc}{b+c} \leq \frac{b+c}{2}, \frac{2ac}{a+c} \leq \frac{a+c}{2}, \frac{2ab}{a+b} \leq \frac{a+b}{2}$$

$$\frac{bc}{b+c} + \frac{ac}{a+c} + \frac{ab}{a+b} \leq \frac{1}{4}(2a + 2b + 2c)$$

$$\frac{bc}{b+c} + \frac{ac}{a+c} + \frac{ab}{a+b} \leq \frac{1}{2}(a + b + c)$$

A04/5/259. Match the following column :

A rational number is given in the form $\frac{a}{b}$; $a, b \in \mathbb{N}$; $\frac{a}{b} \in (0, 1)$ and a and b are co-prime to each other.

Column-I

- (a) $a \cdot b = 10!$ then number of ordered pair (a, b) are less than
- (b) $a \cdot b = 20!$ then number of ordered pair (a, b) are equal to
- (c) $a \cdot b = 30!$ then number of ordered pair (a, b) are greater than
- (d) $a \cdot b = 40!$ then number of ordered pair (a, b) are greater than

Column-II

- (p) 128
- (q) 6
- (r) 621
- (s) 2050
- (t) 500

Choose the correct option :

निम्न कॉलम का सुमेलित कीजिए :

एक परिमेय संख्या $\frac{a}{b}$ के रूप में दी गई है; $a, b \in \mathbb{N}$; $\frac{a}{b} \in (0, 1)$ तथा a तथा b सह-अभाज्य संख्याएँ हैं।

स्तम्भ-I

- (a) $a \cdot b = 10!$ तो क्रमित युग्मों (a, b) की संख्या छोटी होगी
 (b) $a \cdot b = 20!$ तो क्रमित युग्मों (a, b) की संख्या बराबर होगी
 (c) $a \cdot b = 30!$ तो क्रमित युग्मों (a, b) की संख्या अधिक होगी
 (d) $a \cdot b = 40!$ तो क्रमित युग्मों (a, b) की संख्या अधिक होगी

स्तम्भ-II

- (p) 128
 (q) 6
 (r) 621
 (s) 2050
 (t) 500

सही विकल्प का चयन कीजिए :

- (A) $a \rightarrow p$; $b \rightarrow p, r, s, t$; $c \rightarrow p, q, t$; $d \rightarrow p, q, r, t$ (B) $a \rightarrow p$; $b \rightarrow p, q, t$; $c \rightarrow p, r, s, t$; $d \rightarrow p, q, r, t$
 (C) $a \rightarrow p, r, s, t$; $b \rightarrow p$; $c \rightarrow p, q, t$; $d \rightarrow p, q, r, t$ (D) $a \rightarrow p$; $b \rightarrow p, q, t$; $c \rightarrow p, q, r, t$; $d \rightarrow p, r, s, t$

Ans. C

Sol. Any prime factor that occurs in the numerator cannot occur in the denominator. There are 4 prime factors of $10!$. For each of these prime factors i.e. 2, 3, 5, 7. One must decide only whether it occurs in numerator or denominator. Thus 4 decisions can be made in 2^4 ways. They can be grouped into 8 pairs of reciprocals each containing one factor less than 1. (All other cases are similar to this one)

A04/5/260. Five balls are to be placed in three boxes. Each can hold all the five balls. In how many different ways can we place the ball so that no box remains empty, if

Match the following column :

Column-I

- (a) Balls and boxes are all different
 (b) Balls are identical but boxes are different
 (c) Balls are different but boxes are identical
 (d) Balls and boxes are identical

Column-II

- (p) 2
 (q) 25
 (r) 150
 (s) 6

Choose the correct option :

पाँच गेंदों को तीन बक्सों में रखा जाना है। प्रत्येक बक्से में पाँचों गेंदें आ सकती हैं। कितने भिन्न तरीकों से गेंदों को बक्सों में रखा जा सकता है ताकि कोई बक्सा खाली ना रहे, यदि

निम्न कॉलम का सुमेलित कीजिए :

स्तम्भ-I

- (a) बक्से तथा गेंदें भिन्न हैं
 (b) गेंदे समरूप हैं तथा बक्से भिन्न हैं
 (c) गेंदे भिन्न हैं लेकिन बक्से समरूप हैं
 (d) गेंदे तथा बक्से समरूप हैं

स्तम्भ-II

- (p) 2
 (q) 25
 (r) 150
 (s) 6

सही विकल्प का चयन कीजिए :

- (A) $a - r$; $b - s$; $c - p$; $d - q$ (B) $a - r$; $b - p$; $c - s$; $d - q$
 (C) $a - r$; $b - s$; $c - q$; $d - p$ (D) $a - p$; $b - r$; $c - s$; $d - q$

Ans. C

Sol.

- (a) $35 - ({}^3C_1 2^5 - {}^3C_2 1^5) = 150$
- (b) $x_1 + x_2 + x_3 = 5, \quad x_1 \geq 1, x_2 \geq 1, x_3 \geq 1$
 $\Rightarrow {}^{2+3-1}C_{3-1} = 6$
- (c) $\frac{5!}{3!(1!)^2} \times \frac{1}{2!} + \frac{5!}{(2!)^2(1!)} \times \frac{1}{2!} = 25$
- (d) Only 2 cases i.e. (3, 1, 1) and (2, 2, 1)

A04/5/261. $D_1, D_2, \dots, D_{1000}$ are 1000 doors and $P_1, P_2, \dots, P_{1000}$ are 1000 persons. Initially all the doors are closed. P_1 opens all the doors. Then P_2 closes $D_2, D_4, D_6, \dots, D_{998}, D_{1000}$. Then P_3 changes the status of a door means closing it if it is open and opening it if it is closed. Then P_4 changes the status of $D_4, D_8, D_{12}, D_{16}, \dots$ etc. (doors having numbers which are multiple of 4), and so on unit lastly P_{1000} changes the status of D_{1000} .
 Now match the following column :

Column-I

Column-II

- | | |
|---|--------|
| (a) Finally, the number of open doors are | (p) 16 |
| (b) The greatest number of consecutive doors that are closed finally, is | (q) 18 |
| (c) The D_n is door having the greatest number that is finally open, then sum of digits of n is | (r) 31 |
| (d) If D_m is door which is finally closed, where m is greatest 2 digit numbers, then sum of digits of m is | (s) 60 |

Choose the correct option :

$D_1, D_2, \dots, D_{1000}$, 1000 दरवाजे हैं तथा $P_1, P_2, \dots, P_{1000}$, 1000 व्यक्ति हैं। प्रारम्भ में सभी दरवाजे बंद हैं। P_1 सभी दरवाजे खोल देता है। तब $P_2, D_2, D_4, D_6, \dots, D_{998}, D_{1000}$ दरवाजों को बंद कर देता है। तब P_3 एक दरवाजे की स्थिति बदल देता है अर्थात् खुले दरवाजे को बंद कर देता है तथा बंद दरवाजे को खोल देता है। फिर P_4 दरवाजों $D_4, D_8, D_{12}, D_{16}, \dots$ इत्यादि की स्थिति को बदल देता है (दरवाजे जिनका क्रमांक 4 का गुणज है), तथा इसी प्रकार आगे बढ़ते हुए अंत में P_{1000} , दरवाजे D_{1000} की स्थिति बदल देता है।

अब निम्नलिखित स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

- | | |
|--|--------|
| (a) अंत में, खुले दरवाजों की संख्या होगी | (p) 16 |
| (b) अंत में लगातार बंद दरवाजों की अधिकतम संख्या होगी | (q) 18 |
| (c) अंत में D_n वह खुला दरवाजा है जिसकी क्रमांक संख्या अधिकतम है तब n के अंकों का योग होगा | (r) 31 |
| (d) अंत में यदि D_m वह बंद दरवाजा है, जहाँ m, 2 अंकों की अधिकतम संख्या है m के अंकों का योग होगा | (s) 60 |

सही विकल्प का चयन कीजिए :

- | | |
|----------------------------------|----------------------------------|
| (A) $a - q; b - r; c - s; d - p$ | (B) $a - r; b - q; c - s; d - p$ |
| (C) $a - r; b - s; c - q; d - p$ | (D) $a - r; b - s; c - p; d - q$ |

Ans. D

Sol. Any door D_m is operated by person P_n if m is a multiple of n.
 D_m will be finally open, if m has an odd number of factor. And m has an odd number of factors if and only if

m is a perfect square.

$\therefore 1^2, 2^2, 3^2, 4^2, \dots, 31^2$ are the number of the doors that are open finally.

(a) 31 doors are finally open.

(b) D_{900} and D_{961} are largest separated open doors

$\therefore 60$ are largest number of consecutive doors that are closed.

(c) $D_n = D_{961}$, $n = 961$ sum of digits = 16

(d) $D_m = D_{99}$, $m = 99$ sum of digits = 18.

A05/5/262. Match the following column :

Column-I

Column-II

(a) If the binomial coefficients of the r^{th} , $(r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ term in the expansion of $(1+x)^{14}$ are in A.P, then r is equal to (p) 5

(b) The sum of the coefficients in the polynomial expansion of (q) 4

$$\left[(1+x+x^2+x^3+x^4+\dots+(x)^{n-1})(1-x) \right]^m, (n, m \in \mathbb{N}), \text{ is}$$

(c) Sum of the series $\sum_{r=1}^n (-1)^r \cdot {}^nC_r [1+i^r+i^{2r}+\dots+i^{(n-1)r}]$, (r) 0

where $(n=4k, k \in \mathbb{I})$ is equal to (where $i = \sqrt{-1}$)

(d) If $3^{37} = 80\lambda + k$, where $\lambda \in \mathbb{N}$, then k is (s) 3

Choose the correct option :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

स्तम्भ-II

(a) यदि $(1+x)^{14}$ के प्रसार में $r^{\text{वें}}$, $(r+1)^{\text{वें}}$ तथा $(r+2)^{\text{वें}}$ पदों के द्विपद गुणांक समान्तर श्रेणी में हैं, तो r बराबर होगा (p) 5

(b) $\left[(1+x+x^2+x^3+x^4+\dots+(x)^{n-1})(1-x) \right]^m, (n, m \in \mathbb{N})$ के बहुपदीय प्रसार में गुणांकों का योग होगा (q) 4

(c) श्रेणी $\sum_{r=1}^n (-1)^r \cdot {}^nC_r [1+i^r+i^{2r}+\dots+i^{(n-1)r}]$ का योग बराबर होगा, (r) 0

जहाँ $(n=4k, k \in \mathbb{I})$ (जहाँ $i = \sqrt{-1}$)

(d) यदि $3^{37} = 80\lambda + k$, जहाँ $\lambda \in \mathbb{N}$, तो k होगा (s) 3

सही विकल्प का चयन कीजिए :

(A) a-p; b-q; c-r; d-s

(B) a-r; b-p; c-q; d-s

(C) a-p; b-q; c-s; d-r

(D) a-p; b-r; c-r; d-s

Ans. D

Sol. (a) We are given that $2 {}^{14}C_r = {}^{14}C_{r-1} + {}^{14}C_{r+1} \Rightarrow {}^{14}C_{r+1} \Rightarrow 2(r+1)(15-r) = r(r+1) + (15-r)(14-r)$
or $30 + 28r - 2r^2 = r^2 + r + 210 - 29r + r^2$
or $r^2 - 14r^2 + 45 = 0 \Rightarrow (r-5)(r-9) = 0$

$$(b) [(1+x+x^2+x^3+x^4+\dots+(x)^{n-1})(1-x)]^m$$

$$= \left[\left\{ \frac{1-(x)^n}{1-x} \right\} (1-x) \right]^m = (1-(x)^n)^m$$

$$= {}^mC_0 + {}^mC_1(-x)^n + \dots + (-1)^m$$

$${}^mC_m(X)^{mn}$$

So sum of all the coefficients is 0.

$$(c) \sum_{r=1}^n (-1)^r \cdot {}^nC_r \left(\frac{1-i^{nr}}{1-i^r} \right)$$

$$\sum_{r=1}^n \Rightarrow (-1)^r \cdot {}^nC_r \left(\frac{1-i^{4kr}}{1-i^r} \right) = \sum_{r=1}^n (-1)^r \cdot {}^nC_r \left(\frac{1-1}{1-i^r} \right) = 0$$

$$(d) 3^{37} = 3^{4 \times 9} \cdot 3 = 3(81)^9 = 3(80+1)^9$$

$$= 3({}^9C_0 80^9 + {}^9C_1 80^8 + \dots + {}^9C_9)$$

hence remainder is 3.

A05/5/263. For $n \in \mathbb{N}$ and symbols used are having their usual meanings. Match the following column :

Column-I

Column-II

$$(a) \sum_{r=1}^n \binom{n-1}{r-1} \binom{n+1}{n-r+1} \text{ is equal to}$$

$$(p) 2^{n+1}$$

$$(b) ({}^nC_0 \cdot {}^nC_1) + 3({}^nC_1 \cdot {}^nC_2) + 5({}^nC_2 \cdot {}^nC_3) + \dots + (2n-1)({}^nC_{n-1} \cdot {}^nC_n) \text{ is equal to}$$

$$(q) (n+1)2^{n+1}$$

$$(c) 2 \cdot {}^nC_0 + 6 \cdot {}^nC_1 + 10 \cdot {}^nC_2 + \dots + (4n+2) {}^nC_n \text{ is equal to}$$

$$(r) 2^n C_n$$

$$(d) \text{ The sum of coefficient of even powers of } x \text{ in the expansion of } (2x^2+x-1)^{n+2} \text{ is equal to}$$

$$(s) n \cdot 2^n C_{n-1}$$

$$(t) n \cdot 2^n C_{n+1}$$

Choose the correct option :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

$$(a) \sum_{r=1}^n \binom{n-1}{r-1} \binom{n+1}{n-r+1} \text{ का मान होगा}$$

$$(p) 2^{n+1}$$

$$(b) ({}^nC_0 \cdot {}^nC_1) + 3({}^nC_1 \cdot {}^nC_2) + 5({}^nC_2 \cdot {}^nC_3) + \dots + (2n-1)({}^nC_{n-1} \cdot {}^nC_n) \text{ का मान होगा}$$

$$(q) (n+1)2^{n+1}$$

$$(c) 2 \cdot {}^nC_0 + 6 \cdot {}^nC_1 + 10 \cdot {}^nC_2 + \dots + (4n+2) {}^nC_n \text{ का मान होगा}$$

$$(r) 2^n C_n$$

$$(d) (2x^2+x-1)^{n+2} \text{ के प्रसार में } x \text{ की समघातों के गुणांकों का योगफल होगा}$$

$$(s) n \cdot 2^n C_{n-1}$$

$$(t) n \cdot 2^n C_{n+1}$$

सही विकल्प का चयन कीजिए :

$$(A) a \rightarrow s, t; b \rightarrow r; c \rightarrow q; d \rightarrow p$$

$$(B) a \rightarrow s, t; b \rightarrow q; c \rightarrow r; d \rightarrow p$$

(C) $a \rightarrow r; b \rightarrow s, t; c \rightarrow q; d \rightarrow p$

(D) $a \rightarrow s, t; b \rightarrow q; c \rightarrow p; d \rightarrow r$

Ans.

C

Sol.

(a)
$${}^{n-1}C_0 \cdot {}^{n+1}C_n + {}^{n-1}C_1 \cdot {}^{n+1}C_{n-1} + \dots + {}^{n-1}C_{n-1} \cdot {}^{n+1}C_1$$

$$= {}^{n-1+n+1}C_n = {}^{2n}C_n$$

(b)
$$S = ({}^nC_0 \cdot {}^nC_1) + ({}^nC_1 \cdot {}^nC_2) + ({}^nC_2 \cdot {}^nC_3) + \dots + ({}^nC_{n-1} \cdot {}^nC_n)$$

$$S = (2n-1)({}^nC_{n-1} \cdot {}^nC_n) + (2n-3)({}^nC_{n-2} \cdot {}^nC_{n-1}) + \dots + ({}^nC_0 \cdot {}^nC_1)$$

 add

$$\Rightarrow 2S = 2n[{}^nC_0 \cdot {}^nC_1 + {}^nC_1 \cdot {}^nC_2 + \dots + {}^nC_{n-1} \cdot {}^nC_n]$$

$$\Rightarrow 2S = 2n({}^{2n}C_{n-1})$$

$$\Rightarrow S = n \cdot {}^{2n}C_{n-1} = n \cdot {}^{2n}C_{n+1}$$

(c)
$$S = 2 \cdot {}^nC_0 + 6 \cdot {}^nC_1 + 10 \cdot {}^nC_2 + \dots + (4n+2) \cdot {}^nC_n$$

$$S = (4n+2) \cdot {}^nC_n + (4n-2) \cdot {}^nC_{n-1} + \dots + 2 \cdot {}^nC_0$$

$$2S = (4n+4)({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)$$

$$= 4(n+1)2^n$$

$$\Rightarrow S = (n+1)2^{n+1}$$

(d) Let $(2x^2 + x - 1)^{n+2} = a_0 + a_1x + a_2x^2 + \dots + a_{2n+4}x^{2n+4}$
 Replace x by -x

$$\Rightarrow (2x^2 - x - 1)^{n+2} = a_0 - a_1x + a_2x^2 - a_3x^3 + \dots + a_{2n+4}x^{2n+4}$$

 Add both and put x = 1

$$(2x^2 + x - 1)^{n+2} + (2x^2 - x - 1)^{n+2}$$

$$= 2[a_0 + a_2x^2 + \dots + a_{2n+4}x^{2n+4}]$$

$$\Rightarrow 2n+2 + 0 = 2[a_0 + a_2 + a_4 + \dots + a_{2n+4}]$$

$$\Rightarrow a_0 + a_2 + a_4 + \dots + a_{2n+4} = 2^{n+1}$$

A05/5/264. Match the following column :

Column-I

Column-II

(a) The last non-zero digit in the sum $2 \sum_{0 \leq i \leq j \leq 10} i \cdot j {}^{10}C_i {}^{10}C_j$ is (p) 1

(b) If $k = \frac{3n}{2}$, then $\sum_{r=1}^k (-3)^{r-1} {}^{3n}C_{2r-1}$ is, n is even (q) 4

(c) If the expansion of $(1+x+x^2)^n$ be written as (r) 0

$a_0 + a_1x + a_2x^2 + \dots + a_{2n} \cdot x^{2n}$, then the value of $\frac{a_0 + a_1 + a_3 + a_4 + \dots}{a_2 + a_5 + a_8 + \dots}$

(d) The value of $\frac{{}^{47}C_4}{{}^{57}C_4} + \sum_{j=0}^3 \frac{{}^{50-j}C_3}{{}^{57}C_{53}} + \sum_{j=0}^5 \frac{{}^{56-k}C_5}{{}^{57}C_4}$ (s) 2

Choose the correct option :

Match the following column :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

(a) योग $2 \sum_{0 \leq i \leq j \leq 10} i \cdot j^{10} C_i^{10} C_j$ का अंतिम अशून्य अंक होगा

(b) यदि $k = \frac{3n}{2}$, तो $\sum_{r=1}^k (-3)^{r-1} {}^{3n}C_{2r-1}$ होगा, n सम है

(c) यदि $(1+x+x^2)^n$ का प्रसार $a_0 + a_1x + a_2x^2 + \dots + a_{2n} \cdot x^{2n}$

लिखा जा सकता है, तो $\frac{a_0 + a_1 + a_3 + a_4 + \dots}{a_2 + a_5 + a_8 + \dots}$ का मान होगा

(d) $\frac{{}^{47}C_4}{{}^{57}C_4} + \sum_{j=0}^3 \frac{{}^{50-j}C_3}{{}^{57}C_{53}} + \sum_{j=0}^5 \frac{{}^{56-k}C_{53-k}}{{}^{57}C_4}$ का मान होगा

सही विकल्प का चयन कीजिए :

(A) a-p; b-r; c-q; d-s (B) a-r; b-q; c-p; d-s (C) a-q; b-r; c-s; d-p (D) a-r; b-q; c-p; d-s

Ans.

C

Sol.

(a) Let $S = \sum_{0 \leq i \leq j \leq 10} i \cdot j^n C_i^n C_j$

$$= \sum_{i=0}^n \sum_{j=0}^n i \cdot j^n C_i^n C_j - S - \sum_{i=0}^n i^2 \binom{n}{C_i}^2$$

$$\Rightarrow 2S = \sum_{i=0}^n \sum_{j=0}^n i \cdot j^n C_i^n C_j - \sum_{i=0}^n i^2 \binom{n}{C_i}^2$$

$$\Rightarrow 2S = (n2^{n-1})(n2^{n-1}) - \sum_{i=0}^n i^2 \binom{n}{C_i}^2$$

Where, $\sum_{i=0}^n i^2 \binom{n}{C_i}^2$ is the coefficient of x^0 in the product of $n(1+x)^{n-1} n \left(1 + \frac{1}{x}\right)^{n-1}$

Coefficient of x^0 in the product of $\frac{n^2 (1+x)^{2n-2}}{x^{n-1}}$

Coefficient of x^{n-1} in the product $n^2(1+x)^{2n-2} = n^2 {}^{2n-2}C_{n-1}$.

$2S = n^2 \{2^{2n-2} - {}^{2n-2}C_{n-1}\}$ for $n = 10$, we have $2S = 100 \{2^{18} - {}^{18}C_9\} \times 100$

Last digit in 2^{18} is 4 and that the last digit in ${}^{18}C_9$ is zero (0).

Hence last non-zero digit in $2^{18} - {}^{18}C_9$ is 4.

(c) since $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$ _____(1)

Substituting $x = \omega, \omega^2$ and 1 the adding them together

$$a_0 + a_3 + a_6 + \dots = 3^{n-1}$$

Multiplying (i) by x^2 and then repeating the same process again $a_1 + a_4 + a_7 + \dots = 3^{n-1}$

Again multiplying (i) by x and then substituting $x = \omega, \omega^2$ and 1 and then adding the three expansions thus obtained $a_2 + a_5 + a_8 + \dots = 3^{n-1}$

स्तम्भ-II

(p) 1

(q) 4

(r) 0

(s) 2

$$\Rightarrow a_0 + a_3 + a_6 + \dots = a_1 + a_4 + a_7 + \dots = a_2 + a_5 + a_8 + \dots$$

The required ratio is $\frac{2 \cdot 3^{n-1}}{3^{n-1}} = 2$

$$\begin{aligned} \text{(d) Let } E &= \frac{1}{{}^{57}C_4} \left[{}^{47}C_4 + \sum_{j=0}^3 {}^{50-j}C_3 + \sum_{j=0}^5 {}^{56-k}C_{53-k} \right] \\ E &= \frac{1}{{}^{57}C_4} \left[{}^{47}C_4 + ({}^{47}C_3 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3) + \sum_{j=0}^5 {}^{56-k}C_{53-k} \right] \\ \Rightarrow E &= \frac{1}{{}^{57}C_4} \left[{}^{47}C_4 + ({}^{47}C_3 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3) + ({}^{51}C_3 + {}^{52}C_3 + {}^{53}C_3 + {}^{54}C_3 + {}^{55}C_3 + {}^{56}C_3) \right] \\ \Rightarrow E &= \frac{{}^{57}C_4}{{}^{57}C_4} = 1 \end{aligned}$$

A06/5/265. Match the following column :

Column-I

- (a) If A is an idempotent matrix and I is an identity matrix of the same order, then the value of n, such that $(A + I)^n = I + 27$ is
- (b) If $(I - A)^{-1} = I + A + A^2 + \dots + A^7$, then $A^n = O$ where n is
- (c) If A is matrix such that $a_{ij} = (i + j)(i - j)$, then A is singular if order of matrix is
- (d) If a non-singular matrix A is symmetric, show that A^{-1} is also symmetric, then order of A can be

Column-II

- (p) 9
- (q) 10
- (r) 7
- (s) 8

Choose the correct option :

निम्न स्तम्भों को सुमेलित कीजिए :

स्तम्भ-I

- (a) यदि A एक वर्गसम आव्यूह है तथा I एक समान कोटि का तत्समक आव्यूह है तब n का मान होगा, यदि $(A + I)^n = I + 27$
- (b) यदि $(I - A)^{-1} = I + A + A^2 + \dots + A^7$, तो $A^n = O$ जहाँ n है
- (c) यदि A एक आव्यूह इस प्रकार है कि $a_{ij} = (i + j)(i - j)$, तब A अव्युत्क्रमणीय होगा यदि आव्यूह की कोटि है
- (d) यदि एक व्युत्क्रमणीय आव्यूह A सममित है, तो सिद्ध करो कि A^{-1} भी सममित होगा, तो A की कोटि हो सकती है

स्तम्भ-II

- (p) 9
- (q) 10
- (r) 7
- (s) 8

सही विकल्प का चयन कीजिए :

- (A) a-q; a-p; a-r; a-s (B) a-s; a-s; a-p; a-q (C) a-s; a-q; a-r; a-p (D) a-p; a-r; a-q; a-s

Ans. B

Sol. (a) Since A is idempotent, $A^2 = A^3 = A^4 = \dots = A$. Now,

$$\begin{aligned}
(A + I)^n &= I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n \\
&= I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n \\
&= I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n \\
&= I + ({}^nC_1 + {}^nC_2 + \dots + {}^nC_n)A \\
&= I + (2^n - 1)A
\end{aligned}$$

$$\Rightarrow 2^n - 1 = 127 \Rightarrow n = 7$$

(b) We have,

$$\begin{aligned}
&(I - A)(I + A + A^2 + \dots + A^7) \\
&= I + A + A^2 + \dots + A^7 + (-A - A^2 - A^3 - A^4 - \dots - A^8) \\
&= I - A^8 \\
&= I \text{ (if } A^8 = O)
\end{aligned}$$

(c) Here matrix A is skew-symmetric and since $|A| = |A^T| = (-1)^n |A|$, so $|A|(1 - (-1)^n) = 0$. As n is odd, hence $|A| = 0$. Hence A is singular.

(d) If A is symmetric, A^{-1} is also symmetric for matrix of any order.

A08/5/266. Match the following column :

Column-I

Column-II

(a) Consider two events A and B with probabilities of occurrence as

(p) 7

$$P(A) = \frac{3}{4} \text{ and } P(B) = \frac{2}{3} \text{ respectively. If } m \text{ \& } M \text{ denotes minimum}$$

and maximum possible value of $P(A \cap \bar{B})$ respectively, then $\frac{1}{m} + \frac{1}{M}$ is

(b) A pair of dice is rolled and it is known that the obtained sum is

(q) 12

odd. If the probability that the sum is a prime is m, then 9m is equal to

(c) Consider all the permutations of the word MAYAWATI. Number

(r) 15

of words in which all A's occur before Y is to 112k, then k is equal to

(d) Number of ways in which A, B, C, D, E, F can sit on a round table

(s) 20

so that A is neither adjacent to E nor F is equal to 3k, then k is equal to

(t) 21

Choose the correct option :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

(a) माना दो घटनायें A तथा B जिनके होने की प्रायिकता क्रमशः $P(A) = \frac{3}{4}$ तथा

(p) 7

$P(B) = \frac{2}{3}$ है। यदि m तथा M, $P(A \cap \bar{B})$ के क्रमशः न्यूनतम तथा अधिकतम

सम्भव मान है, तो $\frac{1}{m} + \frac{1}{M}$ होगा।

(b) पाँसे के एक युग्म को फेंका जाता है तथा यह ज्ञात है कि प्राप्त योगफल विषम है। यदि

(q) 12

योगफल के अभाज्य होने की प्रायिकता m है, तो 9m का मान होगा

- (c) शब्द MAYAWATI के सभी क्रमचय बनाये जाते हैं। उन शब्दों की संख्या जिनमें सभी A, Y के पहले प्राप्त होते हैं, $112k$ है, तो k का मान होगा (r) 15
- (d) उन तरीकों की संख्या, जिनमें A, B, C, D, E, F को एक गोल मेज पर बिठाया जा सकता है ताकि A ना तो E के आसन्न हो और ना ही F के, $3k$ है, तो k का मान होगा (s) 20
- (t) 21

सही विकल्प का चयन कीजिए :

- (A) $a \rightarrow r; b \rightarrow p; c \rightarrow r; d \rightarrow q$ (B) $a \rightarrow p; b \rightarrow r; c \rightarrow r; d \rightarrow q$
 (C) $a \rightarrow p; b \rightarrow r; c \rightarrow q; d \rightarrow r$ (D) $a \rightarrow p; b \rightarrow q; c \rightarrow r; d \rightarrow r$

Ans.

A

Sol.

(A) $P(A \cap \bar{B}) = P(A) - P(A \cap B)$

$$M = P(A \cap \bar{B})_{\max} = 1 - P(\bar{B}) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$= \frac{3}{4} - \frac{2}{3} = \frac{1}{12}$$

$$\therefore \frac{1}{m} + \frac{1}{M} = 15$$

(B) When n pair of dice is rolled :

Sum is odd in 18 outcomes.

Sum is odd as well as prime in 14 out comes

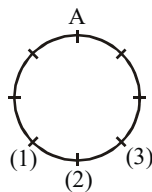
$$\therefore P(\text{odd prime sum}) = m = \frac{14}{18} = \frac{7}{9}$$

$$\therefore 9m = 7$$

(C) ${}^9C_4 \times 1 \times 4! = 112k.$

$$\Rightarrow k = 15$$

(D) 3 possible places for E and F



$$\therefore {}^3C_2 \times 2! \times 3! = 36$$

$$\therefore k = 12$$

A08/5/267. Match the following column :

A box contains n coins. Let $P(E_i)$ be the probability that exactly i out of n coins are biased. If $P(E_i)$ is directly proportional to $i(i+1)$; $1 \leq i \leq n$.

Column-I

Column-II

(a) Proportionality constant k is equal to

(p) $\frac{(3n+1)}{4n}$

- (b) If P be the probability that a coin selected at random is biased then P is (q) $\frac{3}{n(n+1)(n+2)}$
- (c) If a coin selected at random is found to be biased then the probability (r) $\frac{3}{n+2}$
that it is the only biased coin in the box is
- (d) $P(E_n)$ is equal to (s) $\frac{24}{n(n+1)(n+2)(3n+1)}$

Choose the correct option :

सही विकल्प का चयन कीजिए :

निम्न कॉलम का सुमेलित कीजिए :

एक बक्से में n सिक्के हैं। माना $P(E_i)$, n सिक्कों में से ठीक i सिक्कों के पक्षपाती होने की प्रायिकता है। यदि $P(E_i)$, $i(i+1)$; $1 \leq i \leq n$ के समानुपाती है।

स्तम्भ-I

स्तम्भ-II

- (a) समानुपाती नियतांक k का मान होगा (p) $\frac{(3n+1)}{4n}$
- (b) यदि P यादृच्छिक रूप से चुने गए एक सिक्के के पक्षपाती होने की प्रायिकता है तो P का मान होगा (q) $\frac{3}{n(n+1)(n+2)}$
- (c) यदि यादृच्छिक रूप से चुना गया एक सिक्का पक्षपाती पाया जाता है तो प्रायिकता ज्ञात कीजिए कि बक्से में केवल यही एक पक्षपाती सिक्का है (r) $\frac{3}{n+2}$
- (d) $P(E_n)$ बराबर होगा (s) $\frac{24}{n(n+1)(n+2)(3n+1)}$

सही विकल्प का चयन कीजिए :

(A) $a - q, b - p, c - s, d - r$

(B) $a - p, b - q, c - s, d - r$

(C) $a - p, b - s, c - q, d - r$

(D) $a - p, b - s, c - r, d - q$

Ans.

A

Sol.

$$P(E_i) = ki(i+1)$$

$$(a) \quad k \sum_{i=1}^n i(i+1) = 1 \quad k = \frac{3}{n(n+1)(n+2)}$$

$$(b) \quad P(E) = \sum_{i=1}^n P(E_i) \cdot P\left(\frac{E}{E_i}\right) = \sum_{i=1}^n ki(i+1) \cdot \frac{i}{n} = \frac{(2n+1)(n+2)}{4n(n+2)} = \frac{3n+1}{4n}$$

$$P\left(\frac{E_1}{E}\right) = \frac{P(E_1) \cdot P(E/E_1)}{P(E)} = \frac{K \cdot 2 \cdot \frac{1}{2}}{\frac{(3n+1)(n+2)}{4n(n+2)}} = \frac{24}{4n(n+2)}$$